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# An analytical recurrent-steady-state model for inter-rifting deformation at magmatic rifts

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**SUMMARY**
Inter-rifting deformation, the strain that accumulates between discrete magmatic episodes, is the least-constrained phase of the rifting cycle. We model it as the superposed relaxation of a long history of discrete dike intrusions in an elastic plate overlying a Newtonian viscous half-space. The non-local coupling of the plate to the half-space turns post-diking relaxation into a fractional diffusion governed by the half-Laplacian and by a single characteristic velocity $a = EH/[2\eta(1-\nu^2)]$, where $E$, $H$ and $\nu$ are the Young's modulus, thickness and Poisson's ratio of the plate and $\eta$ is the viscosity of the half-space. Deformation spreads from the rift at the constant speed $a$ with a $1/x$ tail, and Fickian diffusion is the same relaxation evaluated at a single diffusion time. Periodic intrusions converge to a recurrent steady state with a closed form in the Digamma and log-Gamma functions. This state conserves the opening of each intrusion over one cycle and has a single length scale, the deformation half-width $aT$, where $T$ is the recurrence interval. Far-field plate divergence therefore emerges without boundary conditions, and steady far-field velocities record the memory of past intrusions rather than continuous creep at depth. A Maxwell visco-elastic half-space with the elastic modulus of the plate modifies this response only within about $1.3\,H$ of the axis and during the first few relaxation times. Applied to post-rifting records from Krafla, Iceland, and Asal, Djibouti, with $T$ derived from the fitted dike opening and a long-term rift opening rate, the model favours a viscous over a visco-elastic half-space and constrains the ratio $EH/\eta$ at both sites. Irregular recurrence, openings and dike positions spread individual velocity profiles around the steady state with little bias.

**Key words:** continental rifting; inter-rifting deformation; fractional diffusion; visco-elastic relaxation; GPS geodesy; Iceland

## 1 INTRODUCTION

Continental rifting occurs at divergent plate boundaries, which drive the transition from stable lithosphere to active oceanic spreading through rifting cycles that span a broad range of spatial and temporal scales (Corti 2009; Ebinger et al. 2013). These cycles alternate between discrete rifting events and the inter-rifting periods that separate them. Each event comprises two phases: elastic strain release through dike emplacement and normal faulting (Grandin et al. 2011; Maccaferri et al. 2016), followed by post-diking visco-elastic relaxation that decays on timescales of years to decades (Foulger et al. 1992; Nooner et al. 2009). Secular plate divergence, set by large-scale mantle dynamics (Saria et al. 2014; Stamps et al. 2021), persists throughout the cycle: it dominates strain accumulation during inter-rifting periods and is superposed on the rapid deformation of rifting events. Geodetic campaigns in Iceland (Heki et al. 1993; Hofton et al. 1995) and the Afar (Vigny et al. 2007; Nooner et al. 2009; Grandin et al. 2011) have documented the two event phases in considerable detail. The mechanisms governing the build-up of extensional strain between discrete events, however, remain the least-constrained aspect of the rifting cycle.

Models of the inter-rifting phase conventionally borrow kinematic frameworks developed for the interseismic earthquake cycle. The standard approach adapts the backslip model of Savage & Prescott (1978): a normal fault bounding the rift is locked from the surface to a depth $H$ and slips continuously at the plate rate below it. The elastic strain that accumulates on the locked segment then reproduces the observed near-field velocity gradient. Alternative formulations represent the diking rift as a tensile dislocation in an elastic half-space that opens continuously at a depth-dependent rate (LaFemina et al. 2005; Konca et al. 2018), or prescribe a basal shear traction at the base of the plate to mimic far-field plate forcing (Ali et al. 2014). These models share

a fundamental physical inconsistency: they ignore the rheological transition at the brittle–ductile boundary of magmatic rifts, where high heat flow and partial melt weaken the lower crust and uppermost mantle (Prodehl et al. 1997; Plasman et al. 2017). Geodetic and seismic constraints place this transition at 5–15 km depth in Iceland and the Afar (Stevens et al. 2021; Weinstein et al. 2017). There, the inferred asthenospheric viscosity ($\eta \approx 1$–$5 \times 10^{18}$ Pa s) implies Maxwell relaxation times of a few years (Hofton & Foulger 1996; Nooner et al. 2009), far shorter than the inter-rifting interval and incompatible with the purely elastic response assumed by backslip models. A continuously opening dislocation is equally problematic: dike intrusions are finite, discrete events, and the geodetic record shows that each episode is followed by a localized transient decaying within a few years (Foulger et al. 1992; Reiss et al. 2022).

Models of the post-diking transient itself meet a related obstacle from two directions. Analytical approaches, exemplified by the diffusion model of Foulger et al. (1992), reproduce the broad deformation pattern after individual intrusions, but they require an a priori viscous layer thickness, and their solutions return to a constant level far from the rift (Figure 1). The observed profiles already capture the transition from post-rifting relaxation to steady inter-rifting divergence. Twelve to seventeen years after the onset of rifting, the displacement field traces a broad, arctangent-like shape rather than a transient confined to the rift, and in the profile of Hofton & Foulger (1996) the two ends of the traverse settle onto distinct levels instead of a common one (Figure 1b). Numerical visco-elastic simulations (Hofton et al. 1995; Hofton & Foulger 1996; Nooner et al. 2009) face an equivalent difficulty: in any finite domain, velocities must vanish at the lateral boundaries, which cannot be reconciled with the steady plate divergence observed in global reference frames (Fernandes et al. 2004; Altamimi et al. 2016). The result is a kinematic gap between the transient post-diking signal and the secular strain rates of global plate motion models (Kreemer et al. 2014; Stamps et al. 2018).

The theoretical foundations laid for cyclic strike-slip fault deformation (Nur & Mavko 1974; Savage & Prescott 1978; Spence & Turcotte 1979; Cohen & Kramer 1984) suggest a different perspective, which we adopt here: inter-rifting deformation is not a distinct physical process but the emergent superposition of visco-elastic responses to an infinite history of discrete dike intrusions. Under this view, the apparent steadiness of inter-rifting velocities arises naturally from the time-averaging of decaying transients, with no need for a continuously creeping source.

We develop an analytical model for an elastic crust overlying a viscous half-space (Hetland 2006) in which dike intrusions recur periodically at interval $T$ and each open the rift by $b$, the dike opening. The governing integro-differential equation reduces, after a Fourier transform, to a fractional diffusion equation (Metzler & Klafter 2000; Samko & Ross 1993) parameterized by a single characteristic velocity $a = EH/[2\eta(1-\nu^2)]$. In the limit of many repeated intrusions, the system reaches a Recurrent Steady State (RSS) expressed exactly through the Digamma and log-Gamma functions, which satisfies the far-field plate divergence constraint by construction. We then extend the model to a Maxwell visco-elastic half-space via the elastic–visco-elastic correspondence principle (Hetland & Hager 2005; Barbot & Fialko 2010), recovering two physically distinct relaxation regimes separated by the Maxwell timescale $\tau_R = \eta/\mu_m$. We apply both models to two geodetic datasets: horizontal displacement profiles from the Krafla rift, Iceland, recorded 12–17 years after the onset of the 1975–1984 rifting sequence (Foulger et al. 1992; Hofton & Foulger 1996), and a 25-year baseline time series together with a rift-perpendicular velocity profile from the Asal rift, Djibouti (Vigny et al. 2007). Together, these records constrain the asthenospheric viscosity of two contrasting settings, show when a geodetic record can distinguish a viscous from a visco-elastic half-space, and show that the decadal post-rifting transient and secular plate divergence are the short- and long-term expressions of a single process.

## 2 ELASTIC-VISCOUS (EV) MODEL

### 2.1 Physical setup and governing equations

The model represents the lithosphere as a coupled mechanical system: a thin elastic plate of thickness $H$ (the seismogenic crust) overlying a Newtonian viscous half-space of viscosity $\eta$ representing the asthenosphere (e.g., Hetland & Hager 2005; Hetland 2006). The horizontal coordinate $x$ is perpendicular to the rift axis; the vertical coordinate $z$ is oriented downward, with the free surface at $z = 0$ and the crust–mantle interface at $z = H$. We consider only the horizontal component of surface displacement; the vertical (uplift) field, although part of the geodetic record (Hofton & Foulger 1996; Nooner et al. 2009), lies outside the scope of this model. Under plane-strain conditions with negligible vertical stress ($\sigma_{zz} \approx 0$), the depth-averaged horizontal stress in the elastic layer is related to the horizontal surface displacement $u(x,t)$ through Hooke's law:

$$\bar{\sigma}_{xx} = \frac{E}{1-\nu^2}\frac{\partial u}{\partial x}, \tag{1}$$

where $E$ is Young's modulus and $\nu$ is Poisson's ratio. The factor $(1-\nu^2)^{-1}$ accounts for the lateral confinement of plane strain. Mechanical equilibrium of the plate requires the horizontal gradient of the depth-averaged stress to be balanced by the shear traction exerted at its base by the underlying viscous mantle:

$$H\frac{\partial \bar{\sigma}_{xx}}{\partial x} + \sigma_{xz}^{\mathrm{BOT}} = 0. \tag{2}$$

The rheological character of the asthenosphere enters through the constitutive relation for the basal traction $\sigma_{xz}^{\mathrm{BOT}}$. For a Newtonian viscous half-space, the shear stress at $x$ depends on the surface velocity at *all* other points through the non-local Stokes kernel (Savage & Prescott 1978):

$$\sigma_{xz}^{\mathrm{BOT}}(x,t) = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2}\,\frac{\partial u}{\partial t}(x',t)\,dx', \tag{3}$$

interpreted as Hadamard finite part. This non-local coupling reflects the long-range character of viscous flow in the half-space: a velocity perturbation at $x'$ decays as $(x-x')^{-2}$ rather than being confined to a finite diffusion radius. In magmatically active rifts, where high heat flow and partial melt weaken the lower crust (Holtzman 2016; Plasman et al. 2017), this coupling operates across the entire plate boundary system.

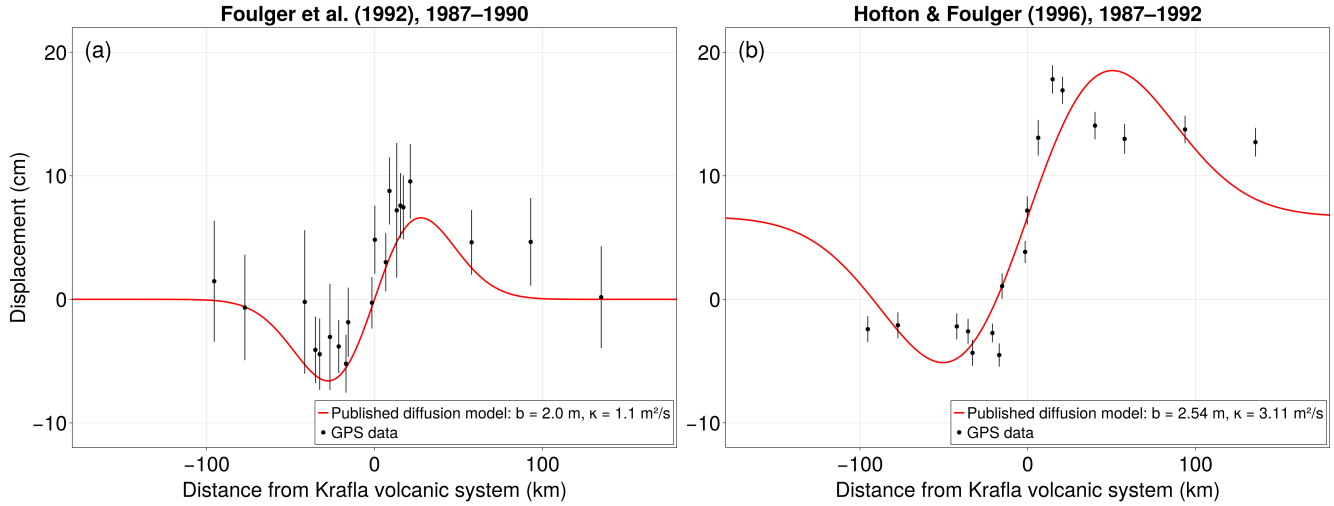



**Figure 1.** Post-rifting displacement profiles across the Krafla rift and the diffusion models published with them, reproduced without refitting. (a) Foulger et al. (1992), their Figure 2b: displacements of 20 stations between 1987 and 1990. (b) Hofton & Foulger (1996), their Figure 5: displacements of 17 stations between 1987 and 1992. Error bars are $1\sigma$. Red curves: the best-fit diffusion model of each study, $u(x,t) = (b/2)\,\mathrm{erfc}[x/(2\sqrt{\kappa t})]$, written here in terms of the full dike opening $b$ (both studies report the dike half-width $U_0 = b/2$). Following Foulger et al. (1992), each profile is modelled as the velocity of this solution at a time $t$ after the intrusions, multiplied by the survey interval, with $t = 11$ yr in (a) and $t = 13$ yr in (b), as each study assumes. Published parameters: (a) $b = 2.0$ m ($U_0 = 1.0 \pm 0.1$ m) and $\kappa = 1.1 \pm 0.3$ m$^2$ s$^{-1}$; (b) $b = 2.54$ m ($U_0 = 1.27 \pm 0.14$ m) and $\kappa = 3.11 \pm 0.68$ m$^2$ s$^{-1}$. The zero reference of profile (b) is arbitrary (Hofton & Foulger 1996), so its curve is shifted by the constant 6.7 cm read from their Figure 16, not fitted. Both curves return to a single constant level far from the rift, whereas the far-field stations of (b) settle on distinct levels on the two flanks, below the curve to the west and above it to the east.

Substituting (1) into (2) and using (3) yields the governing integro-differential equation for $u$:

$$\frac{EH}{1-\nu^2}\frac{\partial^2 u}{\partial x^2} + \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2}\frac{\partial u}{\partial t}(x',t)\,dx' = 0. \qquad (4)$$

Dike intrusions enter as initial conditions: each event opens the rift instantaneously by $b$, offsetting the block on one side of the axis relative to the other (Figure 2). Section 2.3 shows that this offset subsequently redistributes into a symmetric $\pm b/2$ split about the pre-intrusion position far from the rift, while the opening measured at the intrusion site itself remains exactly $b$ at all times.

## 2.2 Fourier reduction and the characteristic velocity

Equation (4) is non-local in $x$ and not tractable in physical space. A spatial Fourier transform resolves this. Defining the transform pair

$$\hat{u}(k,t) = \int_{-\infty}^{+\infty} u(x,t)e^{-ikx}\,dx,$$

$$u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \hat{u}(k,t)e^{ikx}\,dk,$$

the transform of $\partial^2 u/\partial x^2$ is $-k^2\hat{u}$. The transform of the Stokes kernel convolution is:

$$\mathcal{F}\left\{\int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2}\frac{\partial u}{\partial t}\,dx'\right\} = -2\eta|k|\,\frac{\partial \hat{u}}{\partial t}. \qquad (5)$$

Applying the Fourier transform to (4) and using (5):

$$-\frac{EH}{1-\nu^2}k^2\hat{u} - 2\eta|k|\frac{\partial \hat{u}}{\partial t} = 0. \qquad (6)$$

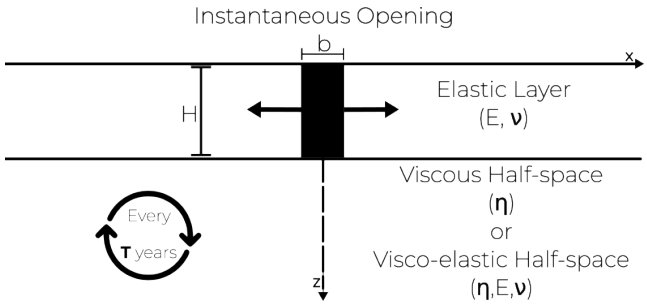



**Figure 2.** Model geometry. The brittle elastic plate (thickness $H$, Young's modulus $E$, Poisson's ratio $\nu$) overlies either a Newtonian viscous half-space (viscosity $\eta$) or, in its visco-elastic extension (Section 3.1), a Maxwell half-space of the same viscosity $\eta$ that additionally shares the plate's elastic moduli $E, \nu$ – the Elastic-Viscous (EV) and Elastic-Visco-Elastic (EVE) cases, respectively. Dike intrusions open the rift by $b$ at intervals $T$; the horizontal surface displacement $u(x,t)$ is positive in the $+x$ direction. The parameter $a = EH/[2\eta(1-\nu^2)]$ sets the characteristic velocity that couples elastic and viscous deformation.

Rearranging and introducing the characteristic velocity

$$a := \frac{EH}{2\eta(1-\nu^2)}, \qquad (7)$$

equation (6) simplifies to a decoupled first-order ODE for each wavenumber $k$:

$$\frac{\partial \hat{u}}{\partial t} = -a\,|k|\,\hat{u}. \qquad (8)$$

The parameter $a$ encodes the full elastic–viscous coupling of the system: large $a$ (stiff crust, low viscosity) implies rapid redistribution of post-diking strain over broad spatial scales;

small $a$ implies slow, locally confined relaxation. In physical space, the operator $|k|$ corresponds to the fractional half-Laplacian $(-\Delta)^{1/2}$, so (8) is equivalently written as the fractional diffusion equation (Metzler & Klafter 2000; Samko & Ross 1993; Samko et al. 1993)

$$\frac{\partial u}{\partial t} = -a\,(-\Delta)^{1/2} u. \tag{9}$$

The solution of (8) for arbitrary initial data is:

$$\hat{u}(k,t) = \hat{u}(k,0)\, e^{-a|k|t}. \tag{10}$$

### 2.3 Step response to a single dike intrusion

The simplest building block of the model is the deformation produced by a single instantaneous dike intrusion at $t = 0$, which opens the rift by $b$. Because the governing equation (4) constrains $u$ only through its gradient, the reference level from which this offset is measured is arbitrary. We therefore write the initial condition, without loss of generality, as a Heaviside displacement, with the symmetric convention $\mathcal{H}(0) := 1/2$ so that the profile is well defined at the rift axis:

$$u(x,0^+) = b\,\mathcal{H}(-x) = \begin{cases} 0\,, & x > 0 \\ b/2\,, & x = 0\,. \\ b\,, & x < 0 \end{cases} \tag{11}$$

Its Fourier transform, evaluated in the distributional sense in Appendix A, is

$$\hat{u}(k,0) = b\pi\delta(k) + \frac{ib}{k}. \tag{12}$$

Substituting (12) into (10) and noting that $\delta(k)$ is active only where $e^{-a|k|t} = 1$:

$$\hat{u}(k,t) = b\pi\delta(k) + \frac{ib}{k}\, e^{-a|k|t}. \tag{13}$$

The inverse transform of (13) reduces to a Dirichlet-type integral (Appendix A). We label quantities of this single-event solution with a STEP superscript, to distinguish them from the periodic Recurrent Steady State (RSS) of §2.4. The step displacement is

$$u^{\mathrm{STEP}}(x,t) = \frac{b}{\pi} \arctan\!\left(\frac{at}{x}\right). \tag{14}$$

Differentiating with respect to $t$, the step velocity is a Lorentzian profile that broadens in time:

$$v^{\mathrm{STEP}}(x,t) = \frac{ba}{\pi}\, \frac{x}{x^2 + a^2 t^2}. \tag{15}$$

Both expressions depend on $x$ and $t$ only through a single combination, $x/(at)$: the self-similarity structure of the step response, developed in §2.3.1 below.

The surface profile is only the boundary trace of a flow that fills the underlying half-space. Depth is reintroduced through the same harmonic extension used to construct the fractional Laplacian in Section 2.2: at depth $z$, the horizontal and vertical interior velocity components are the Poisson-kernel convolution of the surface velocity $V(x)$,

$$\begin{aligned} v_x(x,z) &= \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} \sin^2\theta\, V(x - z\tan\theta)\, d\theta, \\ v_z(x,z) &= \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} \sin\theta\cos\theta\, V(x - z\tan\theta)\, d\theta, \end{aligned} \tag{16}$$

with $V$ set to the EV step velocity $v^{\mathrm{STEP}}(x,t)$ (15).

Figure 3 follows the relaxation of a single intrusion over three orders of magnitude in time, at the surface (left column) and in the underlying half-space (middle and right columns). At the surface, the response is a pulse of divergent velocity that travels away from the rift axis. Its peak sits exactly at $x = a\tau$ (dashed lines), so $a$ is the speed at which post-diking deformation propagates, and its amplitude decays as $b/(2\pi\tau)$. A point at distance $|x|$ has moved by $b/4$, half of its final share of the opening, when the peak passes it at $\tau = |x|/a$; the remaining $b/4$ accumulates only slowly, as $u^{\mathrm{STEP}}$ approaches $b/2$ as $1/\tau$. Ahead of the peak ($|x| \gg a\tau$), Equation (15) reduces to the time-independent tail $ba/(\pi x)$: a distant station records a nearly constant velocity from the moment of intrusion until the peak arrives. Behind the peak ($|x| \ll a\tau$), the velocity falls as $bx/(\pi a\tau^2)$ and vanishes on the axis itself.

The interior field shows the flow that drives this surface pulse. Material rises beneath the axis and turns outward beneath the plate, feeding the divergence of the two flanks (middle column). Because Equation (16) is linear in the surface velocity, the interior field inherits the self-similarity of Equation (15): $\tau$ times the velocity depends only on $x/(a\tau)$ and $z/(a\tau)$. The deforming region therefore deepens at the same rate $a$ as it widens. At distances much larger than $a\tau$ from the intrusion, the flow keeps its early, time-independent pattern, which is why the middle column barely changes across the six rows: even at $\tau = 3\tau_R$, $a\tau \approx 4H$ remains small compared with the $30H$ window. Within $a\tau$, the flow has already slowed down. The enlarged panels (right column) show this as a low-velocity region that opens beneath the axis and grows with $\tau$. The surface itself moves only horizontally ($v_z = 0$ at $z = 0$), consistent with the horizontal-only plate of Section 2.1; the uplift recorded geodetically above rifts lies outside this model.

#### *2.3.1 Self-similarity of the step response*

Both the displacement (14) and velocity (15) of Section 2.3 depend on $x$ and $t$ only through the self-similarity variable $\xi := x/(at)$, which measures distance in units of the velocity-peak position $at$, so that the profile's shape is invariant in $\xi$ at every $t$. Dividing $u^{\mathrm{STEP}}$ by $b$ and $v^{\mathrm{STEP}}$ by $b/t$ collapses both onto universal curves, independent of $E$, $H$, $\eta$, or $b$:

$$\frac{u^{\mathrm{STEP}}}{b} = \frac{1}{\pi} \arctan\!\left(\frac{1}{\xi}\right), \tag{17}$$

$$\frac{t\, v^{\mathrm{STEP}}}{b} = \frac{1}{\pi}\, \frac{\xi}{1 + \xi^2}. \tag{18}$$

Figure 4a plots the displacement collapse (17) directly, alongside the Fickian Foulger et al. (1992) diffusion mode: writing $u_{\mathrm{Foulger}}/b = \frac{1}{2}\,\mathrm{erfc}(\xi_F/2)$ in terms of its own self-similarity variable $\xi_F := x/\sqrt{\kappa t}$, both curves are shown for $\xi > 0$ only. This isolates the decay of the tails, a power law for EV and a Gaussian for Foulger, without reference to any particular parameter.

The same collapse applies to the velocity (18). Differentiating $u_{\mathrm{Foulger}}/b$ in $t$ gives the Fickian model's velocity

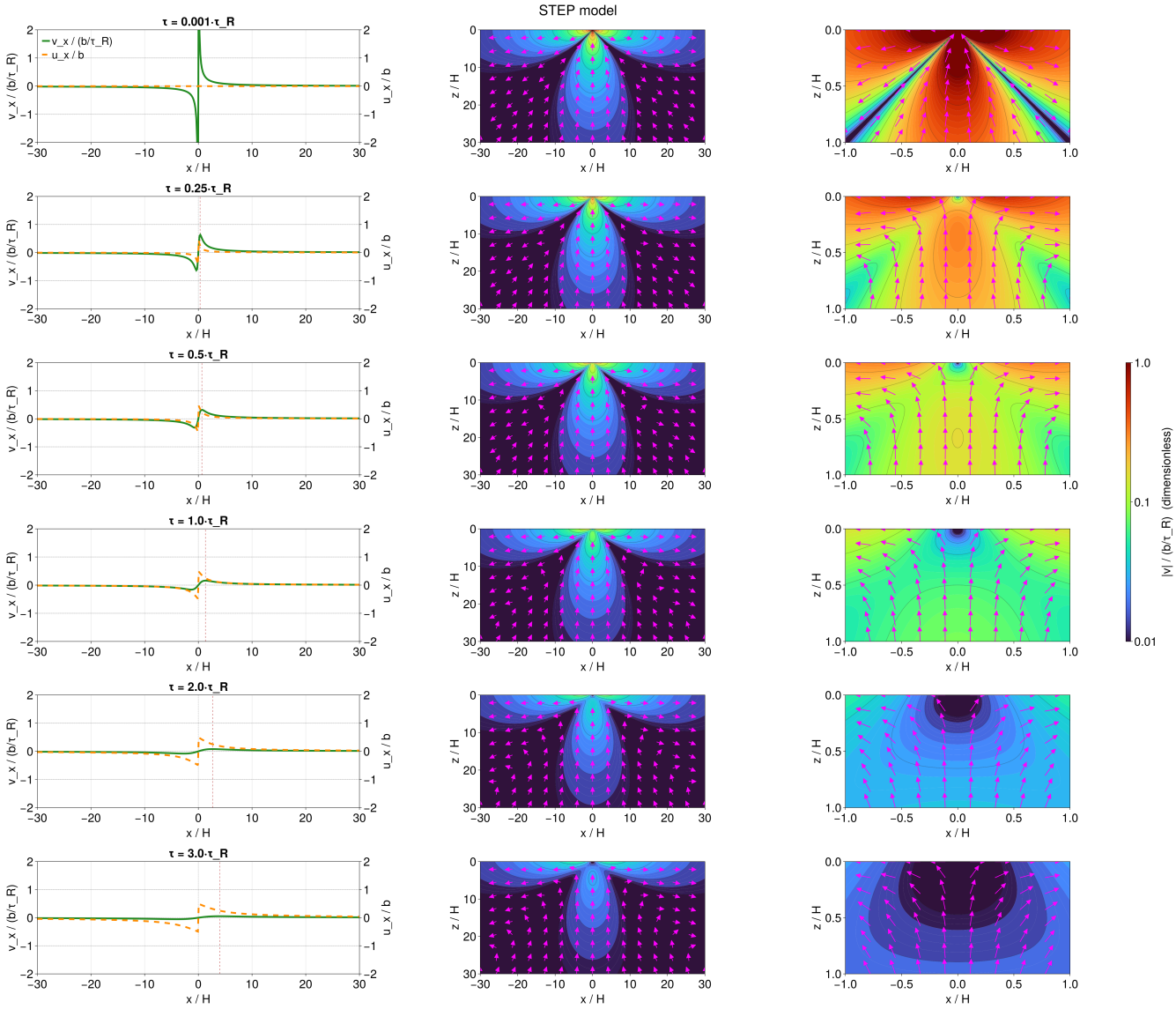



**Figure 3.** EV STEP response to a single dike intrusion of opening $b$ at $t = 0$, at six times $\tau/\tau_R \in \{0.001, 0.25, 0.5, 1, 2, 3\}$, one per row. The Maxwell time $\tau_R = \eta/\mu_m$ (Section 3.1) serves here only as a time unit, the same one used for the EVE counterpart (Figure 8). Left column: surface velocity $v^{\mathrm{STEP}}$ (Equation 15; solid, left axis, in units of $b/\tau_R$) and displacement $u^{\mathrm{STEP}}$ (Equation 14; dashed, right axis, in units of $b$), counted from the instant after the intrusion, so that the two flanks move apart symmetrically and each tends to $\pm b/2$ at the axis. The dashed vertical line marks $x = a\tau$, the position of the velocity peak. Middle column: speed $|v|$ in the half-space beneath the plate (filled contours, logarithmic scale, in units of $b/\tau_R$), computed from the surface velocity with Equation (16) over $|x| \le 30H$ and $z \le 30H$. Arrows show the flow direction only and have a fixed length. Right column: the same field enlarged to $|x| \le H$, $z \le H$ beneath the rift axis, with the same colour scale. Speeds outside the colour bar range [0.01, 1] are clipped to its end colours. All lengths are scaled by the plate thickness $H$. Parameters: $H = 10$ km, $E = 40$ GPa, $\nu = 0.25$, $\eta = 5.2 \times 10^{18}$ Pa s, giving $a \approx 1.3$ km yr$^{-1}$ and $a\tau_R/H = 1/(1-\nu) \approx 1.33$.

collapse:

$$\frac{t\, v_{\mathrm{Foulger}}}{b} = \frac{\xi_F}{4\sqrt{\pi}}\, e^{-\xi_F^2/4}. \tag{19}$$

Figure 4b plots (18) and (19) together: the EV velocity decays as the power law $\xi^{-1}$, a straight line on logarithmic axes, while the Fickian velocity decays as the far steeper $\xi_F\, e^{-\xi_F^2/4}$. Because $\xi$ and $\xi_F$ scale distance by different lengths, $at$ and $\sqrt{\kappa t}$, the horizontal offset between the two curves carries no physical meaning; the robust contrast is the shape of the tail, a power law against a Gaussian.

The two collapses also differ in how the velocity peak moves. The EV peak sits at $\xi = 1$, that is at $x = at$, and migrates outward at the constant speed $a$. The Fickian peak sits at $\xi_F = \sqrt{2}$, that is at $x = \sqrt{2\kappa t}$, and decelerates as $t^{-1/2}$. A station at distance $x$ from the axis therefore records its maximum velocity at $t = x/a$ in the EV model but at $t = x^2/(6\kappa)$ in the Fickian model: the arrival time of the transient grows linearly with distance in the first case and quadratically in the second. Geodetic time series recorded at several distances from the axis can, in principle, discriminate

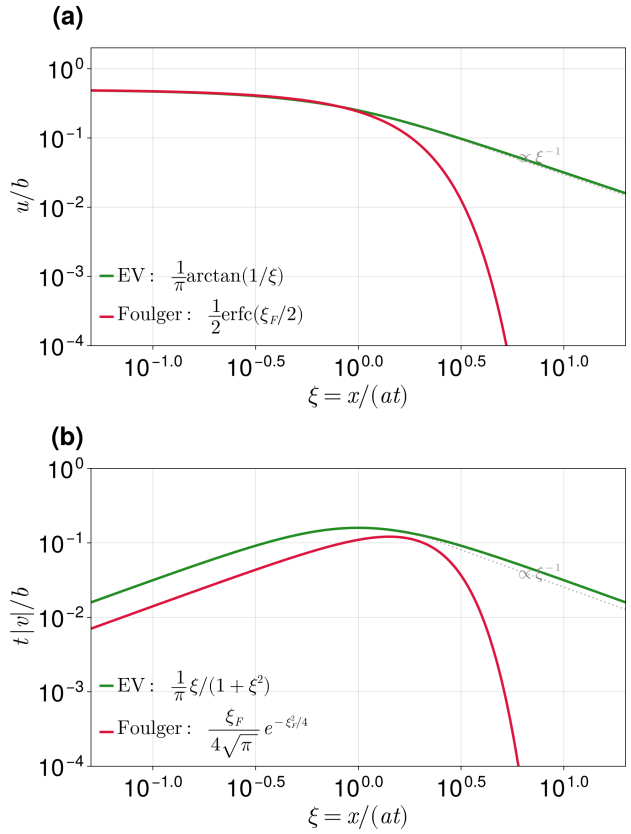



**Figure 4.** Self-similarity collapse of the step response in the EV model and in the Fickian Foulger et al. (1992) model, each plotted against its own self-similarity variable ($\xi = x/(at)$ for EV, $\xi_F = x/\sqrt{\kappa t}$ for Foulger) for $\xi > 0$. Both models start from the same opening $b$, so both displacements equal $b/2$ at the rift axis; the curves are not normalised. (a) Displacement $u/b$: EV (17) and Foulger $\frac{1}{2}\mathrm{erfc}(\xi_F/2)$. (b) Velocity $t\,v/b$: EV (18) and Foulger (19). In both panels the dotted guide marks the $\xi^{-1}$ power-law slope the EV curve approaches; the Fickian curves instead decay as a Gaussian. Because $\xi$ and $\xi_F$ use different length scales, only the shapes of the two tails, not their horizontal offset, are comparable.

between the two on this basis alone, independently of the tail amplitude.

## 2.4 Cyclic accumulation and the Recurrent Steady State

### *2.4.1 Finite sum over past intrusions*

Consider a rift that has hosted $N$ dike intrusions at regular intervals $T$, the most recent a time $\tau \in [0, T)$ ago. Indexing the intrusions by $j = 0, 1, \ldots, N-1$, the number of cycles *before* the most recent one, the $j$-th intrusion has age $\tau + jT$. Superposing the step responses (14), each counted from the instant after its own intrusion, gives the accumulated displacement:

$$u(x,\tau) = \frac{b}{\pi}\sum_{j=0}^{N-1} \arctan\!\left(\frac{a(\tau + jT)}{x}\right). \tag{20}$$

Differentiating with respect to $t$, the corresponding velocity field is:

$$v(x,\tau) = \frac{ba}{\pi}\sum_{j=0}^{N-1} \frac{x}{x^2 + a^2(\tau + jT)^2}. \tag{21}$$

### *2.4.2 The RSS limit and Digamma representation*

After many cycles the system enters a periodic state, the rift analog of the cyclic steady state described for the interseismic earthquake cycle (Thatcher & Rundle 1984; Hetland & Hager 2006). The Recurrent Steady State (RSS) velocity is obtained by extending (21) to an infinite history by letting $N \to \infty$:

$$v_{\mathrm{RSS}}(x,\tau) = \frac{ba}{\pi}\sum_{j=0}^{\infty} \frac{x}{x^2 + a^2(\tau + jT)^2}. \tag{22}$$

Because this series converges absolutely, it reduces to a closed form in the Digamma function $\psi(z)$ (Appendix B):

$$v_{\mathrm{RSS}}(x,\tau) = \frac{b}{\pi T}\,\mathrm{Im}\!\left\{\psi\!\left(\frac{\tau}{T} + i\frac{x}{aT}\right)\right\}. \tag{23}$$

### *2.4.3 RSS displacement and far-field conservation*

Integrating (23) over $\tau$ from the most recent intrusion, and using the antiderivative $d[\ln\Gamma(z)]/dz = \psi(z)$, the RSS displacement accumulated since that intrusion is:

$$u_{\mathrm{RSS}}(x,\tau) = \frac{b}{\pi}\,\mathrm{Im}\!\left\{\ln\Gamma\!\left(\frac{\tau}{T} + i\frac{x}{aT}\right) - \ln\Gamma\!\left(i\frac{x}{aT}\right)\right\}. \tag{24}$$

Over one complete cycle, the cumulative displacement at any fixed $x > 0$ equals half the dike opening:

$$\int_0^T v_{\mathrm{RSS}}(x,\tau)\,d\tau = \frac{b}{2}, \qquad \forall\, x > 0. \tag{25}$$

This is not imposed as a boundary condition but follows analytically from the Digamma representation: $\Gamma(1+iq) = iq\,\Gamma(iq)$ implies $\mathrm{Im}\{\ln\Gamma(1+iq) - \ln\Gamma(iq)\} = \pi/2$, giving the integral $b/2$. The two flanks together therefore separate by $b$ per cycle, exactly the opening of the dike intruded during that cycle, while each flank advances by only half of it. The RSS thus reconciles high-frequency magmatic pulses with secular divergence without unphysical boundary conditions, satisfying the far-field constraint of global plate motion models (Saria et al. 2014; Stamps et al. 2021) by construction.

Figure 5 shows how this conservation plays out over successive cycles. After $n$ complete cycles and a further phase $\tau$, the displacement accumulated since a reference intrusion is $n(b/2)\,\mathrm{sgn}(x) + u_{\mathrm{RSS}}(x,\tau)$. At the end of each cycle, $u_{\mathrm{RSS}}(x, T^-) = b/2$ at every $x > 0$ (Equation 25), so each flank has translated as a rigid block by $b/2$ and the profile is uniform (thick lines in Figure 5a). Within the cycle, the displacement is not uniform: the region next to the axis takes up its share of the opening almost immediately after the intrusion, while distant points catch up later in the cycle. The resulting histories (Figure 5b) range from a staircase at the axis, which jumps by $b/2$ at each intrusion, to a nearly uniform rate $b/(2T)$ far from it. At $x = 0.1\,aT$, nearly 80 per cent of the $b/2$ is gained in the first quarter of the cycle; at $x = aT$, the history departs from the uniform rate by at most $0.04\,b$, and at $x = 3\,aT$ by at most $0.013\,b$. A station at $|x| \gtrsim aT$ therefore records an almost steady velocity, the

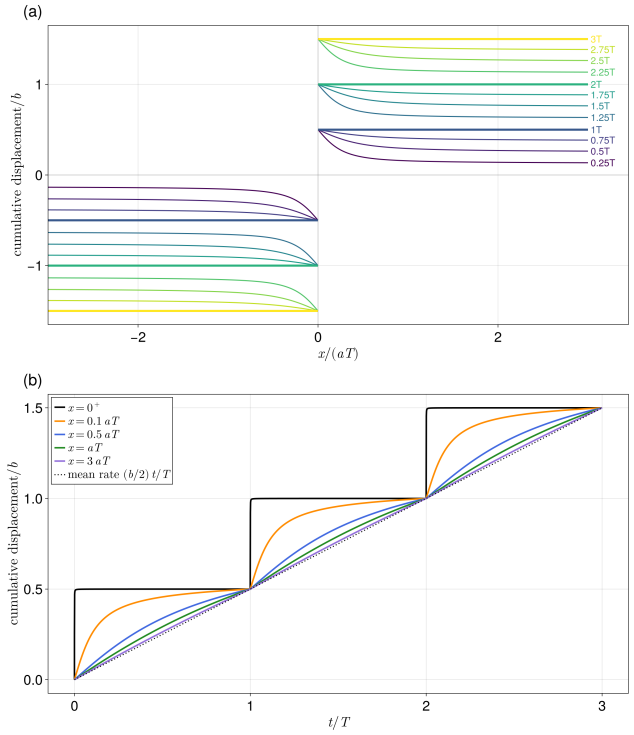



**Figure 5.** Displacement accumulated in the EV RSS over three cycles, starting from an intrusion at $t = 0$ that follows an infinite periodic history. Displacement is normalised by $b$, distance by $aT$ and time by $T$. (a) Profiles every $T/4$; thick lines mark the end of each cycle, when both flanks have moved uniformly by $b/2$ per cycle. (b) Displacement histories at fixed distances from the axis ($x > 0$), from the staircase at $x = 0^+$ to the near-uniform rate at $x = 3\,aT$; the dotted line marks the mean rate $(b/2)\,t/T$.

inter-rifting plate divergence, even though all of the opening is delivered in discrete events.

The interior flow of the RSS follows from the same Poisson-kernel extension (16), with $V$ now set to the EV RSS velocity $v_{\mathrm{RSS}}(x, \tau)$ (23). Figure 6 shows it in the same layout as the STEP field of Section 2.3 (Figure 3), for $T = 5\tau_R$ and at six phases spread over one cycle: the surface profile (left column), the speed in the half-space (middle column), and an enlargement beneath the axis (right column). Distances are scaled by $aT$ and velocities by the rift opening rate $V = b/T$, so colours are not comparable with those of the STEP figure, whose unit is $b/\tau_R$.

The RSS field differs from the STEP field in one essential respect: it never empties out. Far from the axis ($|x| \gg aT$), each flank moves at $V/2$ throughout the cycle, so the two flanks separate at the full rift opening rate $V$, as required by Equation (25). Near the axis, each cycle repeats the behaviour of the single event. Immediately after an intrusion, the near-axis velocity exceeds $V$ by more than an order of magnitude, and a velocity peak migrates away from the axis close to $x = a\tau$ (dashed lines), as in Section 2.3. By the end of the cycle, the near-axis velocity has fallen well below $V/2$, and the profile has smoothed into a monotonic ramp from $-V/2$ to $+V/2$ over a distance of order $aT$. The interior field follows the same pattern. Upwelling beneath the axis and outward flow beneath the plate persist at every phase, and over most of the half-space the speed stays between roughly $0.1\,V$ and $V/2$ throughout the cycle (middle column). Only the region beneath the axis changes: as in the STEP response, a low-velocity region opens there after each intrusion and grows until the next one (right column).

Equation (23) also fixes the scales of the RSS. Normalised by the rift opening rate $V = b/T$, the velocity depends only on the phase $\tau/T$ and on $x/(aT)$, which together form the Digamma argument $\tau/T + ix/(aT)$: $aT$ sets the spatial scale of the deformation zone and $b/T$ its amplitude. Late in the cycle, $aT$ is the distance over which the velocity rises from zero at the axis to the far-field value $\pm V/2$; early in the cycle, the near-axis profile is dominated by the most recent intrusion, whose velocity peak sits at $x \approx a\tau < aT$ (Figure 6). Because $aT \propto EHT/\eta$, a geodetic profile that resolves this scale constrains $EH/\eta$ wherever $T$ can be estimated independently.

The RSS assumes an infinite history of intrusions. Truncating the sum (21) after $N$ intrusions removes the contribution of the older ones; approximating the missing terms by an integral, the truncated velocity falls short of the RSS by $\approx (V/\pi)\arctan[|x|/(NaT)]$. The truncated sum is therefore accurate only within $|x| \ll NaT$ (Figure 7): after ten cycles, the deficit is 3 per cent of $V$ at $|x| = aT$ but 25 per cent at $|x| = 10\,aT$. Physically, the far-field plateau $\pm V/2$ is carried by old intrusions, whose single-event velocity peaks have migrated to $x \approx a(\tau + jT)$; a station at distance $|x|$ records the RSS only if the rift has hosted intrusions for much longer than $|x|/a$. Where this condition fails, as on a newly initiated rift segment, the explicit sum (20) should replace the RSS.

## 3 VISCO-ELASTIC (EVE) EXTENSION

We refer to the two half-space rheologies compared throughout this paper as Elastic-Viscous (EV), for the Newtonian viscous half-space of Section 2, and Elastic-Visco-Elastic (EVE), for the Maxwell visco-elastic half-space developed in this section.

### 3.1 Maxwell half-space and the visco-elastic Stokes kernel

The Maxwell relaxation time $\tau_R := \eta/\mu_m$, where $\mu_m$ is the mantle shear modulus, separates two regimes: on timescales $t \ll \tau_R$ the asthenosphere responds elastically, and on timescales $t \gg \tau_R$ it flows viscously (Bürgmann & Dresen 2008). We model this by replacing the Newtonian half-space with a Maxwell body, a Hookean spring and a Newtonian dashpot in series. By the elastic–visco-elastic correspondence principle (Hetland & Hager 2005; Barbot & Fialko 2010), a Maxwell body responds in the Laplace domain like a dashpot with the frequency-dependent effective viscosity (Appendix C)

$$\tilde{\eta}(s) = \frac{1}{s/\mu_m + 1/\eta} = \frac{\eta}{1 + \tau_R s}. \tag{26}$$

At $s \to 0$ (long times) $\tilde{\eta}(s) \to \eta$, recovering the bare Newtonian dashpot; at $s \to \infty$ (short times) $\tilde{\eta}(s)\,s \to \mu_m$, recovering the purely elastic response of the spring. Replacing $\eta$ by $\tilde{\eta}(s)$ in the Stokes kernel (3) and returning to the time

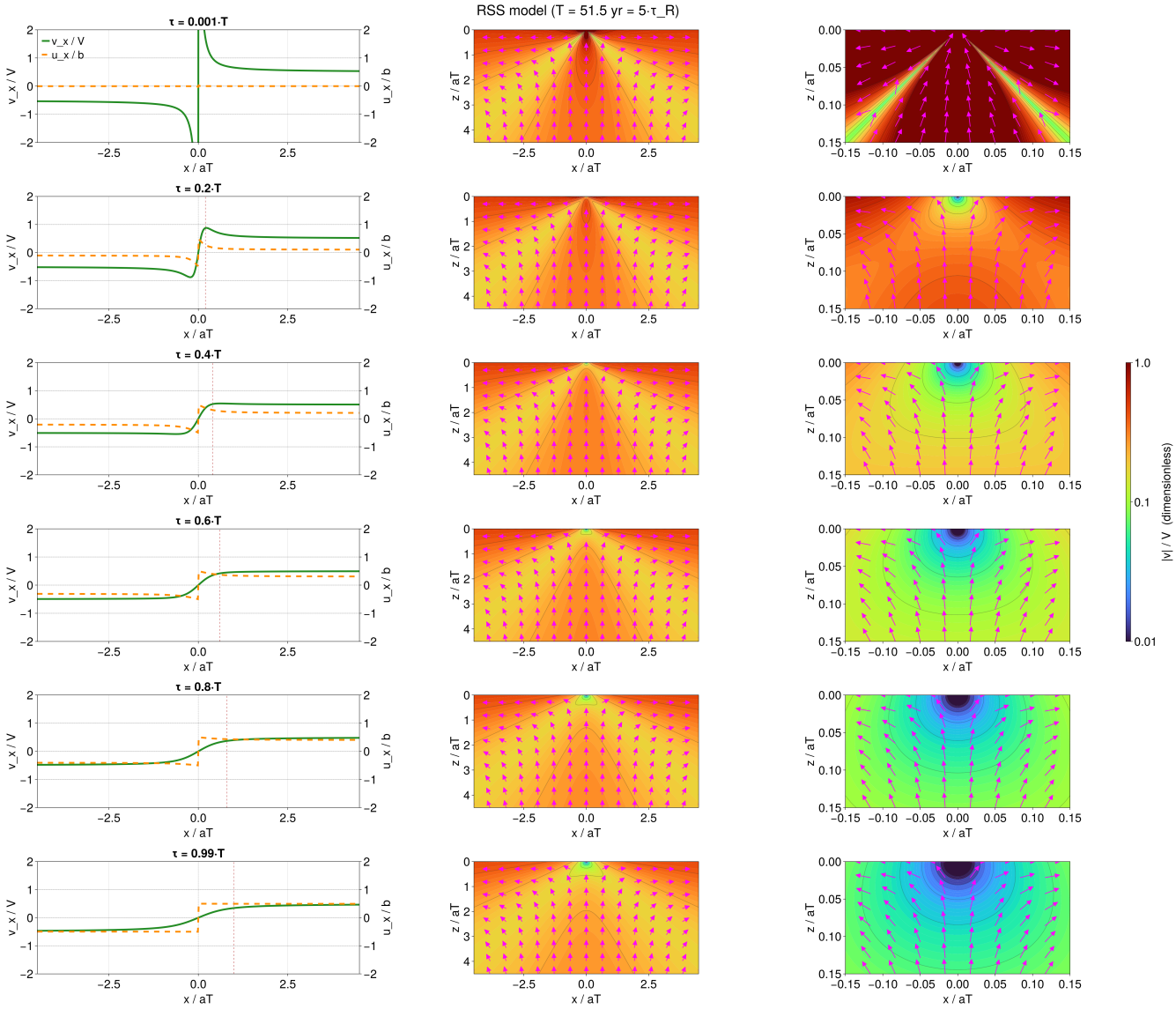



**Figure 6.** EV RSS for $T = 5\tau_R$, in the same layout as the STEP response (Figure 3), at six phases $\tau/T \in \{0.001, 0.2, 0.4, 0.6, 0.8, 0.99\}$, one per row. Left column: surface velocity $v_{\mathrm{RSS}}$ (Equation 23; solid, left axis, in units of $V = b/T$) and displacement accumulated since the most recent intrusion $u_{\mathrm{RSS}}$ (Equation 24; dashed, right axis, in units of $b$). The dashed vertical line marks $x = a\tau$, the velocity-peak position of the most recent intrusion alone. Middle column: speed $|v|$ in the half-space (filled contours, logarithmic scale, in units of $V$), computed with Equation (16) over $|x| \le 4.5\,aT$ and $z \le 4.5\,aT$. Arrows show the flow direction only and have a fixed length. Right column: the same field enlarged to $|x| \le 0.15\,aT$, $z \le 0.15\,aT$ beneath the rift axis, with the same colour scale. Speeds outside the colour bar range [0.01, 1] are clipped to its end colours. Parameters as in Figure 3, giving $aT \approx 67$ km.

domain gives the visco-elastic Stokes kernel (Appendix C):

$$\sigma_{xz}^{\mathrm{BOT}} + \tau_R \frac{\partial \sigma_{xz}^{\mathrm{BOT}}}{\partial t} = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi (x - x')^2} \frac{\partial u(x', t)}{\partial t}\, dx'. \quad (27)$$

In the limit $\tau_R \to 0$, (27) reduces to the EV Stokes kernel (3).

### 3.1.1 *Governing equation in Fourier space*

The equilibrium equation (2) expresses the basal traction in terms of the elastic crust displacement:

$$\sigma_{xz}^{\mathrm{BOT}} = -\frac{EH}{1-\nu^2} \frac{\partial^2 u}{\partial x^2}. \quad (28)$$

Substituting (28) and its time derivative into (27), then applying the spatial Fourier transform using (5):

$$\frac{EH}{1-\nu^2} k^2 \hat{u} + \tau_R \frac{EH}{1-\nu^2} k^2 \frac{\partial \hat{u}}{\partial t} = -2\eta |k| \frac{\partial \hat{u}}{\partial t}. \quad (29)$$

Dividing by $2\eta$, using (7), and collecting $\partial\hat{u}/\partial t$ terms gives the visco-elastic analogue of (8):

$$\frac{\partial \hat{u}}{\partial t} = -\left( \frac{a|k|}{1 + \tau_R a |k|} \right) \hat{u}. \quad (30)$$

The denominator $1+\tau_R a|k|$ acts as a low-pass filter on the relaxation rate, slowing the decay of high-wavenumber (near-field) features relative to the EV case.

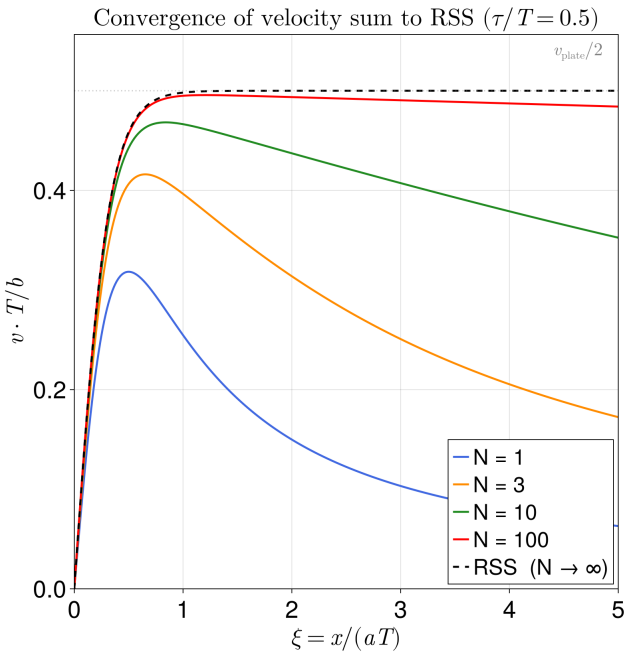



**Figure 7.** Convergence of the finite-sum velocity (21) to the RSS (23) at phase $\tau/T = 0.5$, for $N = 1$, 3, 10 and 2000 past intrusions (solid curves) and the RSS (dashed). Velocity is normalised by $b/T$ and distance by $aT$. The deficit of the truncated sum grows with distance, approximately as $(b/\pi T)\arctan[x/(NaT)]$: with $N = 10$, the sum matches the RSS near the axis but remains below the far-field plateau $b/(2T)$ (dotted) for $x \gtrsim aT$.

### 3.2 Step response and asymptotic behavior

Equation (30) is separable in $k$, yielding a decaying exponential for each wavenumber:

$$\hat{u}(k,t) = \hat{u}(k,0)\exp\left(-\frac{a|k|}{1+\tau_R a|k|}\,t\right). \tag{31}$$

Applying the Heaviside initial condition (12):

$$\hat{u}(k,t) = b\pi\delta(k) + \frac{ib}{k}\exp\left(-\frac{a|k|}{1+\tau_R a|k|}\,t\right). \tag{32}$$

Taking the inverse Fourier transform and following the same parity argument as in Section 2.3, the EVE step displacement requires numerical integration:

$$u^{\text{EVE}}(x,t) = \frac{b}{2} - \frac{b}{\pi}\int_0^{+\infty}\frac{\sin(kx)}{k}\exp\left(-\frac{akt}{1+\tau_R ak}\right)dk, \tag{33}$$

since the rational exponent prevents the closed-form Dirichlet evaluation of Appendix A. The surface velocity follows by differentiating (33) with respect to $t$:

$$v^{\text{EVE}}(x,t) = \frac{ba}{\pi}\int_0^{+\infty}\frac{\sin(kx)}{1+\tau_R ak}\exp\left(-\frac{akt}{1+\tau_R ak}\right)dk. \tag{34}$$

Both integrals are evaluated numerically (Appendix D).

The exponent in (33) reveals two physically distinct regimes. In the long-wavelength limit ($k \to 0$ or $t \to \infty$), $1+\tau_R ak \approx 1$ and the exponential reduces to $\exp(-akt)$, recovering the EV Dirichlet integral of Section 2.3. The elastic memory of the asthenosphere fades at large distances and late times, and the model converges to the Newtonian viscous limit. In the short-wavelength limit ($k \to \infty$ or $t \to 0$), $1+\tau_R ak \approx \tau_R ak$ and the exponent saturates:

$$\lim_{k\to\infty}\left(-\frac{akt}{1+\tau_R ak}\right) = -\frac{t}{\tau_R}. \tag{35}$$

At early times $t \ll \tau_R$, the exponential approaches unity and no near-field deformation occurs outside the rift interface. For $t \sim \tau_R$, near-field deformation decays exponentially at the fixed rate $1/\tau_R$, representing pure Maxwell relaxation of the mantle independent of the elastic crust. Together, (34) and (35) demonstrate that the EVE model introduces a short-time, near-field stiffening relative to the EV model, and that the EVE step response converges to the EV one at $t \gg \tau_R$.

The saturated exponent also gives the near-axis velocity in closed form. For $k \gg 1/(a\tau_R)$, the integrand of (34) reduces to $\sin(kx)\,e^{-t/\tau_R}/(\tau_R k)$, and these wavenumbers control the velocity as $x \to 0$. The Dirichlet integral of Section 2.3 then gives

$$v^{\text{EVE}}(x \to 0^+, t) = \frac{b}{2\tau_R}\,e^{-t/\tau_R}, \tag{36}$$

a finite plateau that decays at the Maxwell rate, whereas the EV velocity vanishes at the axis. The near-field signal of the EVE model is therefore an elastic rebound of the half-space, paced by $\tau_R$ rather than by the diffusion front $at$.

Figure 8 shows the EVE STEP response in the same layout as its EV counterpart (Figure 3): same velocity unit $b/\tau_R$, colour scale, windows, vertical guide $x = a\tau$ and phases, except that its first row shows the limit $\tau \to 0^+$ rather than $\tau = 0.001\,\tau_R$. Any difference between the two figures therefore reflects the rheology alone. Unlike the EV case, the surface velocity is finite immediately after the intrusion and reaches $b/(2\tau_R)$ at the axis, the near-field plateau of Equation (36). Peak surface velocity is largest at $\tau \to 0^+$ and falls with increasing $\tau$, and the underlying flow lobe shrinks and thins in step, the same qualitative decay as the EV STEP field but paced by $\tau_R$ rather than by the diffusion front.

### 3.3 EVE Recurrent Steady State

#### *3.3.1 Infinite sum and Fourier periodicity condition*

The EVE RSS is the superposition of step-response velocities from all preceding intrusions, by the same argument as in Section 2.4:

$$v^{\text{EVE}}_{\text{RSS}}(x,\tau) = \sum_{j=0}^{\infty} v^{\text{STEP,EVE}}(x,\,\tau + jT). \tag{37}$$

Writing $v^{\text{STEP,EVE}}$ in Fourier space (Section 3.1, Equation 34) as $\frac{ba}{\pi}\int_0^{\infty}\frac{\sin(kx)}{1+\tau_R ak}\,e^{-\alpha(k)(\tau+jT)}\,dk$, with the effective Maxwell decay rate

$$\alpha(k) := \frac{ak}{1+\tau_R ak}, \tag{38}$$

each past intrusion contributes a factor $e^{-\alpha(k)jT}$. The series converges for all $k > 0$ and the geometric sum $\sum_{j=0}^{\infty} e^{-\alpha(k)jT} = (1 - e^{-\alpha(k)T})^{-1}$ can be performed inside the Fourier integral. The same result follows from requiring each Fourier mode to repeat from one cycle to

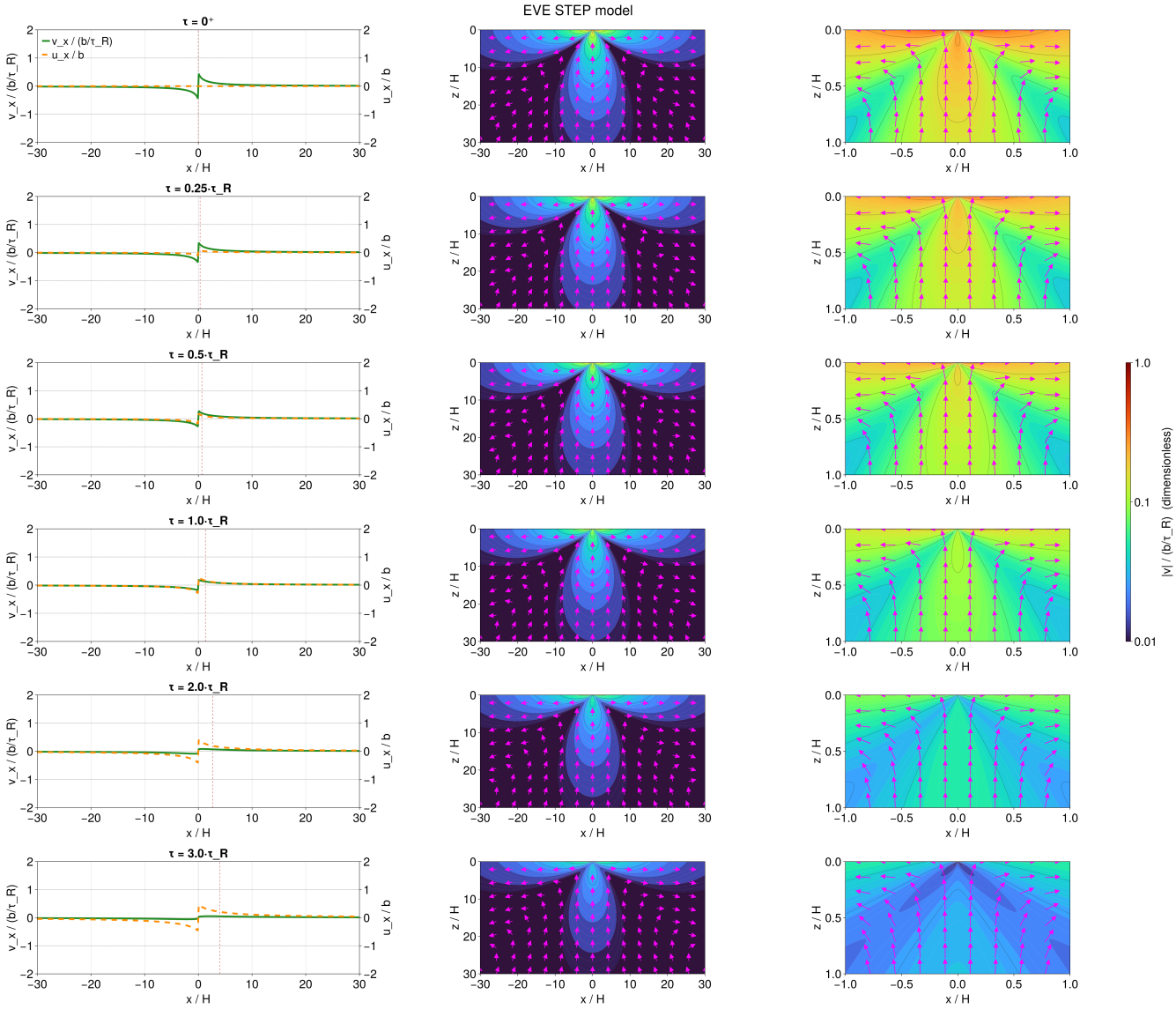



**Figure 8.** EVE STEP response to a single dike intrusion of opening $b$, with the same layout, units, colour scale and parameters as the EV STEP response (Figure 3) and $\tau_R = \eta/\mu_m$ from the shared-modulus closure (Equation 47). Rows show $\tau/\tau_R \in \{0^+, 0.25, 0.5, 1, 2, 3\}$. The first row is the limit $\tau \to 0^+$, evaluated in closed form: unlike the EV case, the surface velocity there is finite, reaching $b/(2\tau_R)$ at the axis (Equation 36). Left column: surface velocity $v^{\mathrm{EVE}}$ (Equation 34; solid, left axis, in units of $b/\tau_R$) and displacement $u^{\mathrm{EVE}}$ (Equation 33; dashed, right axis, in units of $b$), counted from the instant after the intrusion. The dashed vertical line marks $x = a\tau$, the EV velocity-peak position, for direct comparison with Figure 3. Middle column: speed $|v|$ in the half-space (filled contours, logarithmic scale, in units of $b/\tau_R$), computed with Equation (16) over $|x| \le 30H$ and $z \le 30H$. Arrows show the flow direction only and have a fixed length. Right column: the same field enlarged to $|x| \le H$, $z \le H$ beneath the rift axis, with the same colour scale. Speeds outside the colour bar range [0.01, 1] are clipped to its end colours.

the next, apart from the offset added by each intrusion, $\hat{u}(k, 0^+) = \hat{u}(k, T^-) + ib/k$. Together with the decay (30) within the cycle, this condition has the unique solution

$$\hat{u}(k,\tau) = \frac{ib/k}{1 - e^{-\alpha(k)T}}\, e^{-\alpha(k)\tau}, \qquad \tau \in [0, T). \tag{39}$$

### 3.3.2 *Physical-space representation, conservation, and limits*

Inverting (39) gives the EVE RSS velocity and displacement:

$$v^{\mathrm{EVE}}_{\mathrm{RSS}}(x,\tau) = \frac{ba}{\pi}\int_0^\infty \frac{\sin(kx)}{1+\tau_R a k}\,\frac{e^{-\alpha(k)\tau}}{1-e^{-\alpha(k)T}}\,dk, \tag{40}$$

$$u^{\mathrm{EVE}}_{\mathrm{RSS}}(x,\tau) = \frac{b}{\pi}\int_0^\infty \frac{\sin(kx)}{k}\,\frac{1-e^{-\alpha(k)\tau}}{1-e^{-\alpha(k)T}}\,dk. \tag{41}$$

Integrating the velocity over one full period, using $\int_0^T e^{-\alpha\tau}/(1-e^{-\alpha T})\,d\tau = 1/\alpha$ and the algebraic identity

$a/[(1+\tau_R ak)\,\alpha(k)] = 1/k$, gives

$$\int_0^T v_{\rm RSS}^{\rm EVE}(x,\tau)\,d\tau = \frac{b}{\pi}\int_0^\infty \frac{\sin(kx)}{k}\,dk = \frac{b}{2}, \qquad (42)$$

where the last step is the Dirichlet integral. Half the dike opening, $b/2$, is recovered every cycle, independent of $\tau_R$, matching the EV RSS result (25) and the late-time limit $b/2$ of the single-event EVE displacement (33). In the limit $\tau_R \to 0$, $\alpha(k) \to ak$ and $(1+\tau_R ak)^{-1} \to 1$, and (40) reduces to the EV RSS Fourier integral whose closed form is the Digamma series of Section 2.4. For $\tau_R > 0$, the saturating denominator $1 + \tau_R ak$ prevents this reduction and (40) is evaluated numerically (Appendix D).

The interior flow of the EVE RSS again follows from Equation (16), with $V$ set to $v_{\rm RSS}^{\rm EVE}$ (40). Figure 9 shows it in the same layout, normalisation and phases as the EV RSS (Figure 6), for the same $T = 5\tau_R$, except that its first row shows the limit $\tau \to 0^+$. Far from the axis ($|x| \gg aT$), each flank moves at $V/2$ throughout the cycle, as in the EV RSS. Near the axis, the cycle starts from a finite velocity jump at the source. Each past intrusion contributes the plateau (36) at its own age $\tau + jT$, and summing these contributions gives

$$v_{\rm RSS}^{\rm EVE}(x \to 0^+,\,\tau) = \frac{b}{2\tau_R}\,\frac{e^{-\tau/\tau_R}}{1-e^{-T/\tau_R}}, \qquad (43)$$

so the velocity jumps by twice this value across the axis for any $\tau_R > 0$, whereas the EV RSS vanishes continuously there. Immediately after an intrusion, the near-axis velocity is about $2.5\,V$ (Equation 43), the elastic rebound of the most recent intrusion together with the remnants of earlier ones. This jump decays at the rate $1/\tau_R$, to about $0.9\,V$ at $\tau = 0.2\,T = \tau_R$ and $0.3\,V$ at $\tau = 0.4\,T$. By the end of the cycle it has all but vanished, and the profile has smoothed into a monotonic ramp from $-V/2$ to $+V/2$ over a distance of order $aT$, as in the EV RSS. The interior field follows the same pattern as in the EV case. Upwelling beneath the axis and outward flow beneath the plate persist at every phase, and the speed over most of the half-space stays between roughly $0.1\,V$ and $V/2$ (middle column). A low-velocity region opens beneath the axis after each intrusion and grows until the next one (right column).

### 3.4 Constraining $\tau_R$ without an extra free parameter

Section 4 fits the EVE RSS to real data alongside its EV counterpart. A free $\tau_R$ alongside $(a, b)$ would let the extra parameter absorb noise rather than test whether the EVE model improves the fit: a model with one more free parameter always reaches at least as low a $\chi^2$, whether or not that parameter is physically warranted. We instead derive $\tau_R$ from $a$, so that the EVE RSS carries exactly as many free parameters as the EV RSS it is compared against.

Doing so requires the shear modulus $\mu_m$ of the half-space, left general in Equation (26) so far. By definition, $\tau_R := \eta/\mu_m$ is a material property of the half-space alone. The EV half-space is purely viscous and has no elastic modulus of its own: its decay rate $a|k|$ (8) depends on wavenumber and is not a material relaxation rate. The analogous timescale of the coupled plate–half-space *system* is the same viscosity-over-modulus ratio, with the shear modulus of the plate in place of that of the half-space:

$$\tau_{\rm EV} := \frac{\eta}{\mu_{\rm plate}}, \qquad \mu_{\rm plate} := \frac{E}{2(1+\nu)}. \qquad (44)$$

$\tau_R$ and $\tau_{\rm EV}$ therefore differ only in which shear modulus, $\mu_m$ or $\mu_{\rm plate}$, is paired with the half-space viscosity $\eta$.

We adopt the simplest closure available without additional data: the half-space shares the same shear modulus as the elastic plate it underlies,

$$\mu_m = \mu_{\rm plate} := \frac{E}{2(1+\nu)}. \qquad (45)$$

This closure makes the two relaxation times identical by construction, with no free parameter left to tune the agreement:

$$\tau_R = \frac{\eta}{\mu_m} = \frac{\eta}{\mu_{\rm plate}} = \tau_{\rm EV}. \qquad (46)$$

Combined with $\eta = EH/[2a(1-\nu^2)]$, the inversion of the characteristic velocity (7), this gives

$$\tau_R = \tau_{\rm EV} = \frac{H}{a(1-\nu)}, \qquad (47)$$

a relaxation time set entirely by the fitted characteristic velocity $a$ and the canonical parameters $H$ and $\nu$ (Section 5.4), with no independent estimate of $\mu_m$ or $\eta$. The shared-modulus closure is a simplifying assumption, not a physical necessity: the mantle shear modulus need not equal that of the crust. It is, however, the natural default in the absence of independent constraints, and it keeps the EV-versus-EVE comparison of Section 4 an equal-parameter test of functional form.

## 4 APPLICATIONS TO MAGMATIC RIFT SYSTEMS

We test the EV and EVE Recurrent Steady States (Sections 2.4 and 3.3) against post-diking geodetic records from two magmatic rifts. The first is the 1975–1984 Krafla rifting episode in NE Iceland, sampled by two rift-perpendicular displacement profiles (Foulger et al. 1992; Hofton & Foulger 1996). The second is the 1978 Asal–Ghoubbet episode in the Afar Triangle, sampled by a 25-year baseline time series and a rift-perpendicular velocity profile (Vigny et al. 2007). The two sites differ in rift opening rate, in the length of their records and in the type of data, so together they test whether a single functional form describes inter-rifting deformation in contrasting settings.

Both records cover only part of one cycle. They span 12–17 yr after the onset of the Krafla episode and 1–25 yr after the Asal event, and in neither case has a later intrusion been documented. We set the recurrence interval through the far-field conservation of the RSS (Equation 25). Each flank advances by $b/2$ per cycle, so the two flanks separate by the full dike opening $b$ every $T$, and the rift opening rate, the rate at which the two flanks separate, is $V = b/T$, with each flank moving at $\pm V/2$ far from the axis. We take $V$ from an independent, long-term estimate of the rift opening rate, from global plate motion models or regional geodetic studies, and derive the recurrence interval from the fitted opening as $T = b/V$. This leaves two free parameters,

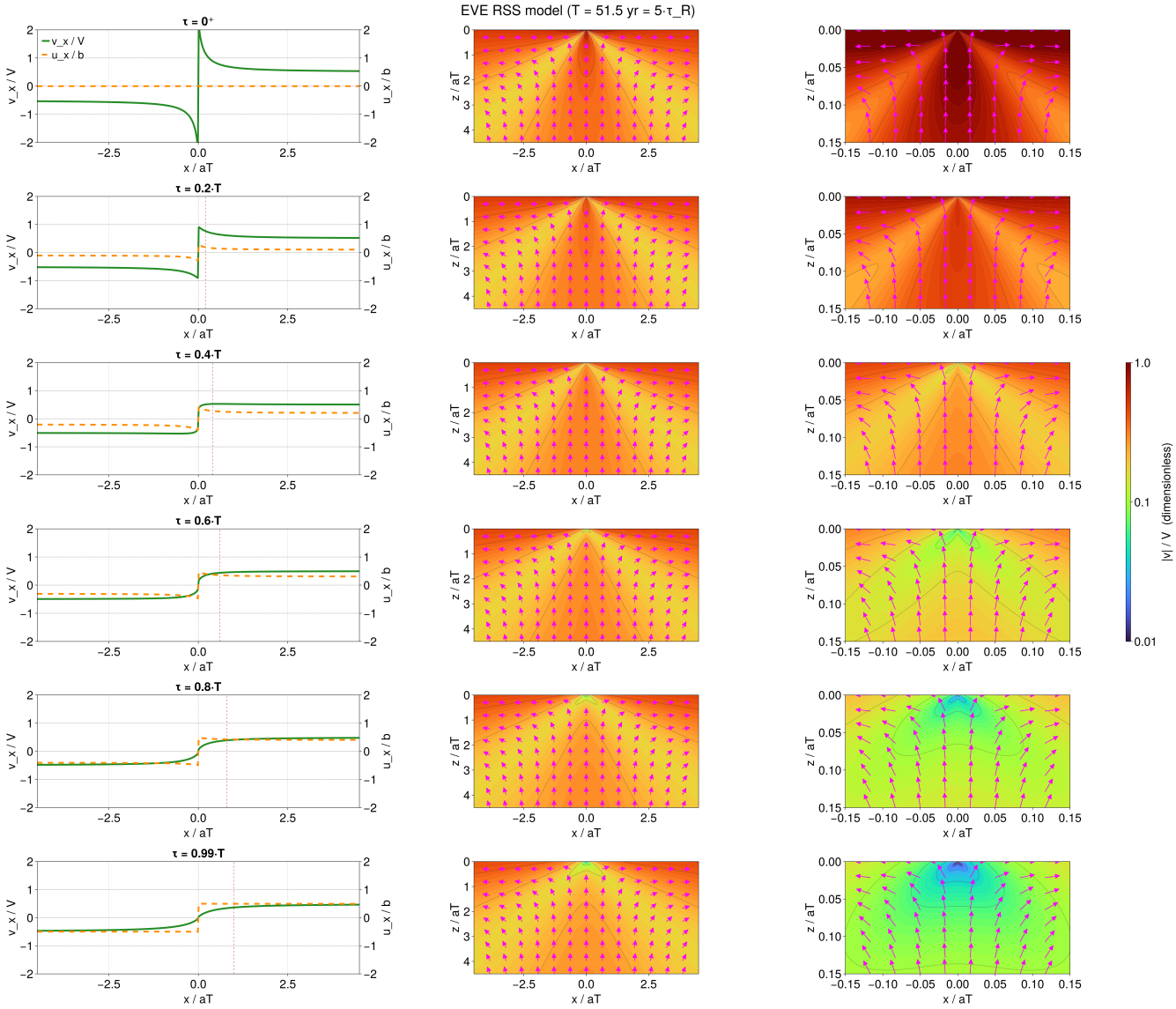



**Figure 9.** EVE RSS for $T = 5\tau_R$, in the same layout, units and colour scale as the EV RSS (Figure 6), with $\tau_R = \eta/\mu_m$ from the shared-modulus closure (Equation 47), at six phases $\tau/T \in \{0^+, 0.2, 0.4, 0.6, 0.8, 0.99\}$, one per row. The first row is the limit $\tau \to 0^+$, computed from its exact form: the displacement since the most recent intrusion is zero, and the surface velocity at the axis, $(b/2\tau_R)/(1 - e^{-T/\tau_R}) \approx 2.5\,V$ (Equation 43), lies beyond the plotted range and is clipped at $2\,V$. Left column: surface velocity $v^{\mathrm{EVE}}_{\mathrm{RSS}}$ (Equation 40; solid, left axis, in units of $V = b/T$) and displacement accumulated since the most recent intrusion $u^{\mathrm{EVE}}_{\mathrm{RSS}}$ (Equation 41; dashed, right axis, in units of $b$). The dashed vertical line marks $x = a\tau$, the EV velocity-peak position of the most recent intrusion alone, for direct comparison with Figure 6. Middle column: speed $|v|$ in the half-space (filled contours, logarithmic scale, in units of $V$), computed with Equation (16) over $|x| \le 4.5\,aT$ and $z \le 4.5\,aT$. Arrows show the flow direction only and have a fixed length. Right column: the same field enlarged to $|x| \le 0.15\,aT$, $z \le 0.15\,aT$ beneath the rift axis, with the same colour scale. Speeds outside the colour bar range [0.01, 1] are clipped to its end colours. Parameters as in Figure 3, giving $aT \approx 67$ km.

the characteristic velocity $a$ and the dike opening $b$, plus a reference-frame offset where the data require one. The EVE RSS carries $\tau_R$ derived from $a$ through the shared-modulus closure (Equation 47), so each EVE fit has exactly as many free parameters as its EV counterpart. Although the records are short, the RSS still contains the contribution of all earlier intrusions, including the far-field plate motion they sustain.

At each site we minimise the misfit

$$\chi^2 = \sum_{i=1}^{N} \left( \frac{u_i^{\mathrm{obs}} - u_i^{\mathrm{pred}}}{\sigma_i} \right)^2 , \qquad (48)$$

where $u_i^{\mathrm{obs}}$ and $u_i^{\mathrm{pred}}$ are the observed and predicted displacement, baseline change or velocity at data point $i$, $\sigma_i$ is its $1\sigma$ uncertainty, and $N$ is the number of observations. We report the reduced statistic $\chi^2_r = \chi^2/(N-p)$, where $p$ is the number of free parameters, so that $\chi^2_r \approx 1$ indicates a fit within measurement noise. We compare the EV and EVE fits with the Akaike information criterion (AIC). Because both models carry the same number of free parameters, the AIC difference reduces to the difference in $\chi^2$, and any preference reflects the shape of the predicted deformation alone.

**Table 1.** Best-fit RSS parameters for the joint Krafla and Asal fits, with $1\sigma$ ranges in brackets ($\Delta\chi^2 \leq 2.30$ projections, two parameters). The recurrence interval is derived as $T = b/V$ ($V = 1.8$ cm yr$^{-1}$ at Krafla, 13 mm yr$^{-1}$ at Asal) and $\tau_R$ from $a$ through (47); $c$ is the reference-frame offset of the Hofton & Foulger (1996) profile, and $k$ the number of free parameters. At Asal, $\chi^2/N$ is 1.37 (EV) and 1.62 (EVE) on the 33 time-series points and 21.3 and 25.4 on the 15 profile points. $\Delta$AIC is relative to the EV RSS at the same site.

| Site | Model | $a$ ($10^{-5}$ m s$^{-1}$) | $b$ (m) | $T$ (yr) | $\tau_R$ (yr) | $c$ (cm) | $k$ | $\chi^2_r$ | $\Delta$AIC |
|---|---|---|---|---|---|---|---|---|---|
| Krafla ($N = 37$) | EV RSS | 4.99 [4.21, 5.82] | 1.71 [1.60, 1.80] | 95 [89, 100] | – | 6.5 | 3 | 1.61 | – |
| | EVE RSS | 11.9 [10.8, 13.1] | 1.60 [1.51, 1.69] | 89 [84, 94] | 3.6 [3.2, 3.9] | 6.6 | 3 | 1.82 | +7.1 |
| Asal ($N = 48$) | EV RSS | 1.00 [0.97, 1.04] | 0.673 [0.666, 0.681] | 52 [51, 52] | – | – | 2 | 7.93 | – |
| | EVE RSS | 6.95 [6.77, 7.14] | 0.642 [0.637, 0.648] | 49 [49, 50] | 6.1 [5.9, 6.2] | – | 2 | 9.46 | +70.3 |

### 4.1 Krafla rift, Iceland

The Krafla rifting episode began in December 1975 and lasted about a decade, with some 20 dike intrusions along the Krafla fissure swarm (Hofton & Foulger 1996). Its aftermath was recorded by GPS campaigns in 1987, 1990 and 1992. We digitized two rift-perpendicular displacement profiles from these campaigns. The first comprises 20 stations of Foulger et al. (1992) (their Figure 2b), differenced between 1987 and 1990. The second comprises 17 stations of Hofton & Foulger (1996) (their Figure 5, the same profile they compare with the diffusion model in their Figure 16), differenced between 1987 and 1992. Each point carries its published $1\sigma$ error bar. The two profiles also differ in their spatial reference. Foulger et al. (1992) tie their displacements to near-rift stations that showed no significant deformation, whereas Hofton & Foulger (1996) state that the zero level of their profile is arbitrary. We therefore add a constant offset $c$ to the model for the Hofton & Foulger (1996) profile alone and solve for it during the fit.

A joint fit first requires a common time origin for the two profiles. Our RSS represents each episode as a single intrusion of opening $b$, and it predicts each profile from the elapsed times $(\tau_i, \tau_f)$ since that intrusion, so both datasets must measure these times from the same instant. Neither published study provides one directly: both measure time from an origin that differs from the onset of the episode. Foulger et al. (1992) treat their 1987–1990 displacements as an instantaneous velocity 11 yr after the dike intrusions, and Hofton & Foulger (1996) treat their 1987–1992 displacements as one 13 yr after, so both place the intrusion in 1979. Hofton & Foulger (1996) likewise inject all dikes of their visco-elastic model simultaneously in 1979, and report that spreading the injections over the real nine-year sequence changes the 1987–1992 field negligibly. We adopt the onset of the episode, $t_0 = 1975$, for both datasets. The half-space begins to relax as soon as the first dike opens the elastic plate, in December 1975. An origin in 1979 would treat the stresses imposed by the intrusions of the preceding four years as if their relaxation had not yet started. Because the opening was in fact spread over the episode, $t_0 = 1975$ gives the longest elapsed time compatible with the record, and the times since intrusion used below are upper bounds. The two epoch pairs then sit at $\tau = 12$–$15$ yr and 12–17 yr after the intrusion.

With this common origin, we fit both profiles jointly, because they record the same episode. A single $(a, b)$ is shared by the two profiles, each is evaluated at its own $(\tau_i, \tau_f)$, and the pooled $\chi^2$ (48) is minimized over the $N = 37$ stations combined, with $c$ as a third free parameter. The recurrence interval follows from $T = b/V$ with $V = 1.8$ cm yr$^{-1}$, the NUVEL-1A full spreading rate for north Iceland (DeMets et al. 1994), which Hofton & Foulger (1996) adopt as the time-averaged opening rate across the region. Foulger et al. (1992) used the 1.9 cm yr$^{-1}$ predicted by the earlier NUVEL-1 model, which the NUVEL-1A revision of the geomagnetic time scale superseded.

Figure 10 and Table 1 summarize the result. Both models reproduce the near-rift step and the far-field plateau of the two profiles (Figure 10a,b) with $\chi^2_r < 2$, and the data favour the EV RSS at the same parameter count ($\Delta$AIC $= +7.1$).

Both $\chi^2$ landscapes are closed and single-minimum within the $3\sigma$ contour (Figure 10c,d), so $a$ and $b$ are resolved independently despite the short observation window (Table 1). In the EVE landscape $a$ and $b$ are positively correlated, while in the EV landscape they are nearly independent. Section 5.3 discusses why the two rheologies are hard to separate at Krafla, and Section 5.4 the implied viscosity and recurrence interval.

### 4.2 Asal rift, Djibouti

The November 1978 Asal–Ghoubbet rifting episode in the Afar Triangle opened approximately 2 m of rift-perpendicular extension over several days (Le Dain et al. 1979; Ruegg et al. 1984). Vigny et al. (2007) tracked the EP00–DF00 baseline, a rift-perpendicular geodetic tie crossing the Asal rift, from 1979 to 2003. The baseline lengthened by $\approx 0.5$ m, rapidly at first and much more slowly after the mid-1980s. Vigny et al. (2007) also measured rift-perpendicular GPS velocities along a transect crossing the rift, at a mean epoch of $\approx 19$ yr after the event. We digitized all 33 time-series points from their Figure 3 and all 15 profile points from their velocity transect, and fit both jointly with the EV and EVE RSS forward models of Sections 2.4 and 3.3. The predicted baseline change is $\Delta L = 2u(x_0, \tau)$ at $x_0 = 2000$ m, half the EP00–DF00 rift-perpendicular separation, with the rifting onset fixed at $t_0 = 1978.85$ yr (Vigny et al. 2007). The predicted velocity profile is evaluated at the transect epoch, with the model centre placed at $x = +1$ km northeast of the topographic rift axis, following Vigny et al. (2007). Because the Vigny et al. time series is referenced to the first measurement epoch ($\tau_{\mathrm{ref}} \approx 1.1$ yr after the rifting event) rather than to the rifting onset, we subtract from each model curve its own prediction at $\tau_{\mathrm{ref}}$, anchoring $\Delta L = 0$ at the reference epoch without introducing a free parameter.

As at Krafla, the recurrence interval follows from the

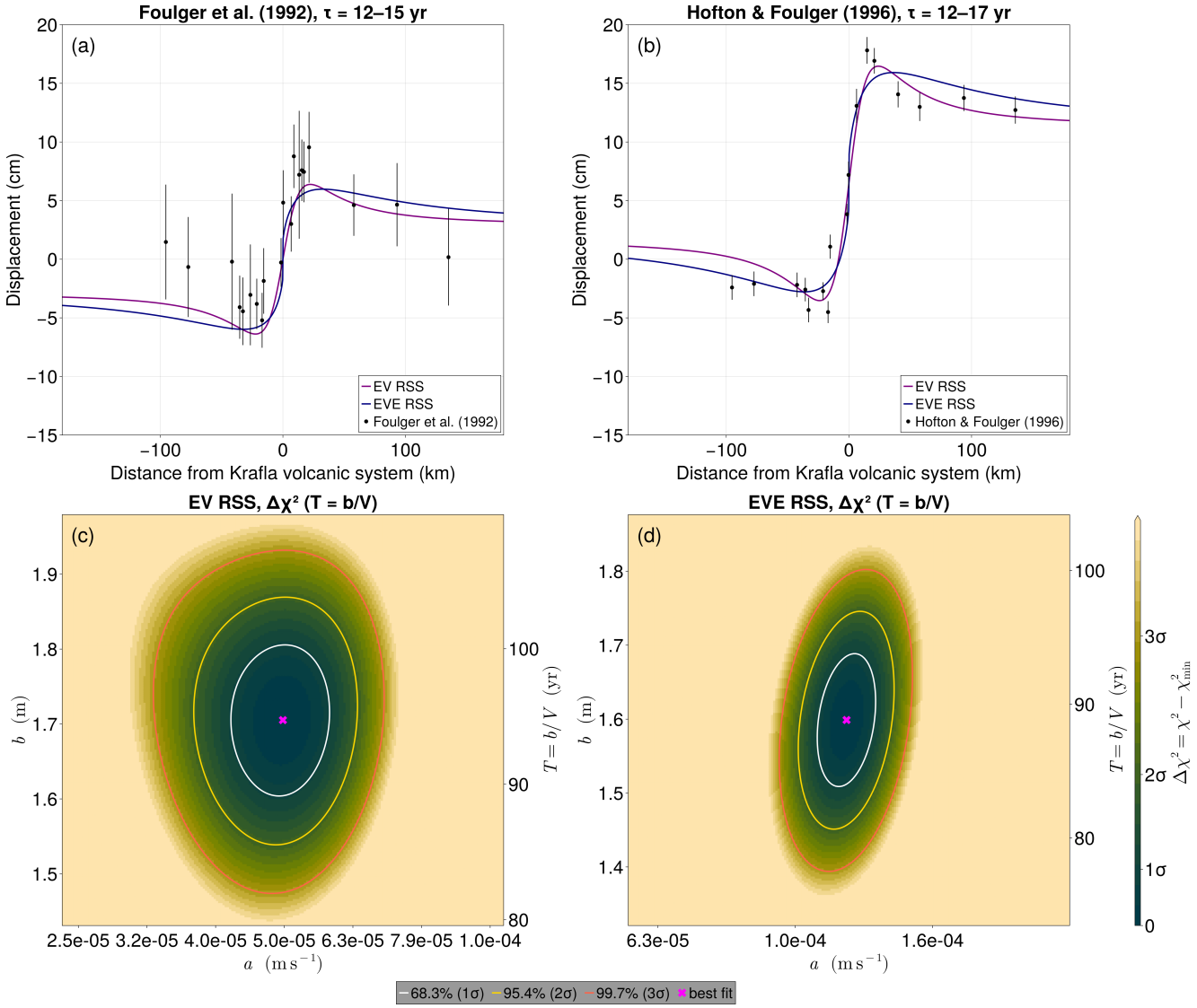



**Figure 10.** Joint EV and EVE RSS fits to both Krafla datasets, with $T$ derived from the fitted dike opening $b$ through the opening-rate relation $T = b/V$ at $V = 1.8$ cm yr$^{-1}$ and time measured from $t_0 = 1975$. (a) Foulger et al. (1992) ($\tau_i = 12$ yr, $\tau_f = 15$ yr). (b) Hofton & Foulger (1996) ($\tau_i = 12$ yr, $\tau_f = 17$ yr), plotted at its own uncorrected displacements, with the reference-frame offset absorbed into each curve rather than removed from the data. Curves: EV RSS (purple) and EVE RSS (navy), each with one $(a, b)$ shared by both panels; error bars are $1\sigma$. (c, d) Joint $\chi^2$ landscapes in $(a, b)$ for the EV and EVE RSS, shown as $\Delta\chi^2 = \chi^2 - \chi^2_{\min}$, with the reference-frame offset minimised at every grid point. Contours mark the 68.3, 95.4 and 99.7 per cent confidence levels for two parameters ($\Delta\chi^2 = 2.30$, 6.17, 11.83); the cross marks the best fit. The right axis gives the derived $T = b/V$. See Table 1 for goodness-of-fit statistics.

opening-rate relation $T = b/V$. We take $V = 13$ mm yr$^{-1}$, the long-term Arabia–Somalia rate reported by Vigny et al. (2007). We do not use the $\approx 16$ mm yr$^{-1}$ opening they measure across the Asal rift itself: they attribute its excess over the long-term rate to the post-1978 transient, which the RSS already produces, so adopting it as $V$ would count the transient twice. The velocity profile is expressed in the Somalia reference frame, in which the SW flank is nearly fixed and the NE flank moves at the full rate. Because the model's far-field asymptotes are $\pm V/2$, the frame shift is $V/2$ by construction rather than a fitted offset. No second intrusion has been documented at Asal–Ghoubbet since 1978: Smittarello et al. (2016) analyse over 40 yr of geodetic monitoring (1972–2016) as the relaxation of that single event. We therefore require $T \geq T_{\min} = 37.2$ yr, the 1978–2016 span, which under $T = b/V$ becomes the lower bound $b \geq V\,T_{\min} \approx 0.48$ m. Only $(a, b)$ remain free, and the EVE variant carries $\tau_R$ derived from $a$ through (47).

Figure 11 and Table 1 summarize the result. Neither fit sits on the $T_{\min}$ floor. Both models reproduce the time series closely, with $\chi^2/N = 1.37$ for the EV RSS and 1.62 for the EVE RSS over its 33 points. The elevated joint $\chi^2_r$ comes from the profile ($\chi^2/N = 21.3$ and 25.4 over its 15 points), where the observed SW–NE asymmetry exceeds what a single symmetric transient can produce. The EVE profile also shows a velocity jump of $\approx 4.6$ mm yr$^{-1}$ at the model centre (Figure 11b), the RSS jump of Equation (43) at $\tau \approx 19$ yr. The EV RSS fits better at equal parameter count, with

$\Delta$AIC = +70.3 in its favour. Section 5.3 discusses why the Asal record separates the two rheologies, and Section 5.4 the implied viscosity and the field observations that the fits do not reproduce.

Figure 11c,d shows the joint (time series + profile) $\chi^2$ landscapes for both rheologies. The combined dataset confines $(a, b)$ to a narrow, anti-correlated ellipse in both cases, well above the $T_{\min}$ floor, with $1\sigma$ ranges of a few per cent (Table 1).

## 5 DISCUSSION

### 5.1 Inter-rifting deformation as the memory of discrete intrusions

Kinematic models of inter-rifting deformation need a free parameter that the mechanics of a magmatic rift does not supply. Backslip models (Savage & Prescott 1978) and their dislocation equivalents (LaFemina et al. 2005; Konca et al. 2018) reproduce observed velocity gradients through a locking depth or a creep-zone geometry, which carries no independent constraint where the rheological transition is gradational. Numerical visco-elastic simulations face the same obstacle at their lateral boundaries: these must either force the far-field velocity to zero, in contradiction with global plate motion models (Altamimi et al. 2016), or prescribe the divergence rate that the model is meant to predict (Hofton et al. 1995; Hofton & Foulger 1996).

The RSS removes both requirements, because it treats inter-rifting deformation as the superposed relaxation of all past intrusions rather than as a separate process. Far-field plate divergence follows from the conservation of the opening over each cycle (Equation 25), not from a boundary condition, and the only length scale is the deformation half-width $aT = EHT/[2\eta(1-\nu^2)]$, set by the balance between elastic stresses in the plate and viscous drag at its base. A station that records a steady velocity far from the rift therefore does not reveal continuous creep at depth: it records the cumulative relaxation of intrusions that occurred many cycles earlier.

This memory sets the conditions under which the RSS applies. Truncating the history after $N$ intrusions leaves a velocity deficit of about $(V/\pi)\arctan[|x|/(NaT)]$ (Section 2.4), so the RSS is reproduced to within 5 per cent of the far-field velocity $V/2$ only where $N \gtrsim 13\,|x|/(aT)$. The number of intrusions a rift must have hosted thus grows linearly with distance from its axis, a direct consequence of the slowly decaying $1/x$ tail of the step response (Section 2.3.1). The farthest stations used here lie at $0.9\,aT$ at Krafla and $0.6\,aT$ at Asal, so the RSS describes them provided each rift has hosted more than about 12 and 8 intrusions, about 1100 and 400 yr of rifting at the derived recurrence intervals. On a newly initiated rift segment, or at stations far beyond $aT$, the explicit finite sum (20) should replace the RSS.

### 5.2 Non-local versus Fickian relaxation

The EV model and the diffusion model of Foulger et al. (1992) differ in the order of their governing equation, not only in their parameters. The Fickian model, $\partial u/\partial t = \kappa\,\partial^2 u/\partial x^2$, damps each Fourier mode at the rate $\kappa k^2$; the EV model (Equation 8) damps it at $a|k|$, the Fourier signature of the half-Laplacian. This operator arises from the $1/(x-x')^2$ Stokes kernel (3): the half-space transmits stress instantaneously, so every point of the plate feels an intrusion from the first instant, and no diffusive front is needed to reach it. The step velocity is therefore a Lorentzian with a $1/x$ tail (18), against the Gaussian decay of the Fickian velocity (19, Figure 4b).

The two solutions are nevertheless exactly related. By Bochner subordination (Metzler & Klafter 2000), the EV decay factor is an average of Fickian decay factors over an effective diffusion time $s$,

$$e^{-a|k|t} = \int_0^\infty e^{-k^2 s}\, g_t(s)\, ds, \qquad g_t(s) = \frac{at}{2\sqrt{\pi}}\, s^{-3/2} \exp\!\left(-\frac{a^2t^2}{4s}\right), \tag{49}$$

so the EV step displacement (14) equals the Fickian profile $(b/2)\,\mathrm{erfc}[x/(2\sqrt{s})]$ averaged over $s$ with the weight $g_t(s)$. We checked this identity numerically to a relative precision better than $10^{-7}$. The weight peaks at $s = (at)^2/6$ but decays only as $s^{-3/2}$, so long effective diffusion times contribute substantially, and these carry the $1/x$ tail. The Fickian model retains a single diffusion time, $s = \kappa t$, and discards that tail.

The two formulations leave distinct, observable signatures. The EV velocity peak migrates at the constant speed $a$, so a station at distance $x$ records its peak at $t = x/a$; the Fickian peak arrives at $t = x^2/(6\kappa)$ (Section 2.3.1). The far fields also differ: the RSS settles onto the plate velocity $\pm V/2$, while the Fickian profile returns to its reference level. Krafla illustrates the consequence. We refit the diffusion model of Figure 1 under the same conditions as the RSS: one $(\kappa, b)$ shared by the two profiles, $t_0 = 1975$, and the offset $c$ of the Hofton & Foulger (1996) profile solved during the fit. The refit gives $\kappa = 2.70$ m$^2$ s$^{-1}$ and $b = 2.81$ m with $\chi^2_r = 4.90$, against 1.61 for the EV RSS at equal parameter count. The refit diffusion curves peak $\approx 50$ km from the axis, whereas the data peak at 15–21 km, and they return to their reference level within $\sim 150$ km. Networks that span both the near-rift peak and the far-field plateau, or time series at several distances from the axis, are therefore best placed to discriminate between the two formulations.

### 5.3 Viscous versus visco-elastic half-space: when the data can tell

At equal $a$, the EVE model departs from the EV model only at short wavelengths and early times. The elastic response of the Maxwell half-space saturates the decay rate of modes with $k \gtrsim 1/\ell_M$, where $\ell_M = a\tau_R$ (Equation 35), and its effect fades after a few $\tau_R$. Its signature is a finite velocity at the axis that decays as $e^{-t/\tau_R}$ (Equation 36), which in the RSS becomes a velocity jump across the axis (Equation 43). This jump follows from treating the dike as infinitely thin and the plate as a single layer; in reality we expect it to be spread over a distance comparable to the plate thickness $H$. Under the shared-modulus closure, $\ell_M = H/(1-\nu)$ (Equation 47), about 13 km for $H = 10$ km, independently of $a$: the elastic signal is confined within a distance of about $1.3\,H$ from the axis.

Whether a record can distinguish the two rheologies therefore depends on two numbers: its first epoch in units

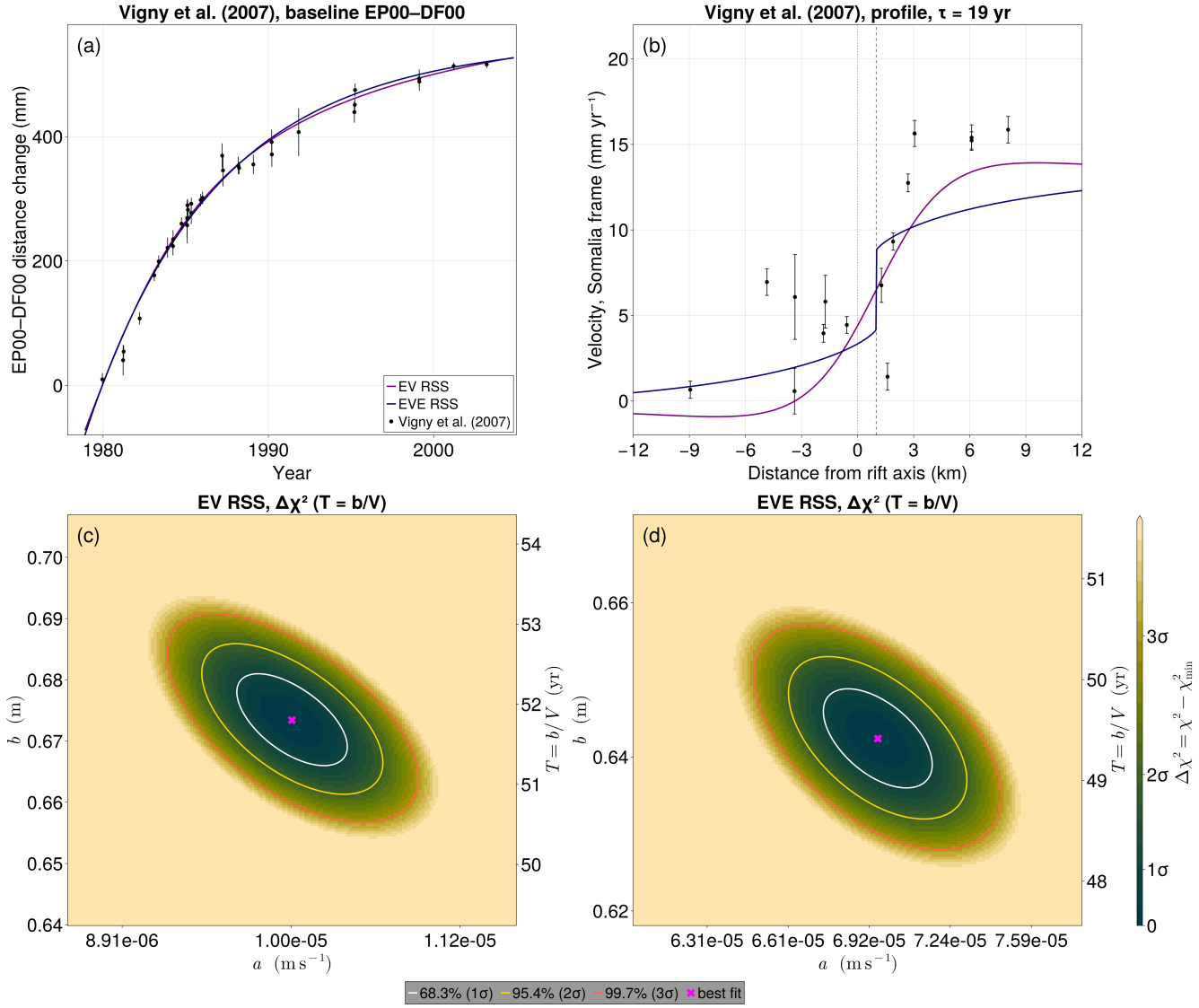



**Figure 11.** Joint EV and EVE RSS fits to the Asal data of Vigny et al. (2007), with $T$ derived from the fitted dike opening $b$ through the opening-rate relation $T = b/V$ at $V = 13$ mm yr$^{-1}$. (a) EP00–DF00 baseline change (mm), filled circles with $1\sigma$ error bars; model curves are referenced to the first measurement epoch ($\tau_{\mathrm{ref}} \approx 1.1$ yr). (b) Rift-perpendicular velocity profile at $\tau \approx 19$ yr in the Somalia reference frame (mm yr$^{-1}$); model curves include the frame shift $V/2$. The dotted line marks the topographic rift axis and the dashed line the model centre at $x = +1$ km. Curves: EV RSS (purple) and EVE RSS (navy), each with one $(a, b)$ shared by both panels. (c, d) Joint $\chi^2$ landscapes in $(a, b)$ for the EV and EVE RSS, shown as $\Delta\chi^2 = \chi^2 - \chi^2_{\mathrm{min}}$. Contours mark the 68.3, 95.4 and 99.7 per cent confidence levels for two parameters ($\Delta\chi^2 = 2.30$, 6.17, 11.83); the cross marks the best fit. The right axis gives the derived $T = b/V$. See Table 1 for goodness-of-fit statistics.

of $\tau_R$, and the fraction of its stations that lie within $\ell_M$. At Krafla, the observations start $3.4\,\tau_R$ after the onset, and only five of 20 and three of 17 stations lie within $\ell_M \approx 13$ km of the axis. Most of the elastic contribution has decayed by then, and the data favour the EV RSS only modestly ($\Delta\mathrm{AIC} = +7.1$). At Asal, the time series starts at $0.2\,\tau_R$ and the whole velocity profile lies within $\ell_M$. This record samples the window where the two rheologies differ most, and the comparison is decisive ($\Delta\mathrm{AIC} = +70.3$ in favour of the EV RSS).

Two caveats qualify this preference. First, it rests on the shared-modulus closure, which ties $\tau_R$ and $\ell_M$ to $a$ and $H$; a mantle shear modulus different from that of the plate would shift both scales. Second, the choice of rheology changes the inferred viscosity. The EVE fits require a larger characteristic velocity than the EV fits, by a factor of 2.4 at Krafla and 6.9 at Asal (Table 1), which lowers $\eta$ to $1.8 \times 10^{18}$ and $3.1 \times 10^{18}$ Pa s at $H = 10$ km and $E = 40$ GPa. At Asal, this rheological uncertainty exceeds the $\chi^2$ uncertainty on $a$ by more than an order of magnitude. Viscosities inferred from post-diking geodesy are therefore conditional on the assumed half-space rheology as much as on the elastic parameters.

### 5.4 What the fits constrain, and where the model falls short

The fits constrain the composite parameter $a = EH/[2\eta(1-\nu^2)]$, not the viscosity alone. For fixed $E$, the best-fit $a$ maps onto an iso-$a$ band in the $(H, \eta)$ plane (Figures 12 and 13), and an independent estimate of the seismogenic thickness, $H \approx 5$–$15$ km in Iceland and the Afar (Stevens et al. 2021; Plasman et al. 2017; Weinstein et al. 2017), turns it into a viscosity. For the canonical $H = 10$ km, $E = 40$ GPa and $\nu = 0.25$, the EV RSS gives $\eta \approx 4.3 \times 10^{18}$ Pa s at Krafla and $\eta \approx 2.1 \times 10^{19}$ Pa s at Asal. The $1\sigma$ ranges from the $\chi^2$ landscapes alone are narrow, $[3.7, 5.1] \times 10^{18}$ and $[2.05, 2.21] \times 10^{19}$ Pa s. Adding an uncertainty on $E$ over [20, 80] GPa in quadrature widens them to $[2.1, 8.7] \times 10^{18}$ and $[1.1, 4.3] \times 10^{19}$ Pa s: the elastic modulus, not the fit, dominates the error budget. The Krafla value lies below the bound $\eta < 10^{19}$ Pa s of Sigmundsson (1991) and within the post-glacial rebound range for Iceland (Sigmundsson et al. 2010), consistent with the hot sub-lithospheric mantle of the Iceland plume (Holtzman 2016). The Asal value is five times higher, and higher than estimates for the Dabbahu segment of the same rift system ($\eta \approx 5 \times 10^{18}$ Pa s; Nooner et al. 2009; $\eta \sim 10^{18}$ Pa s for $H = 21$ km; Hamling et al., 2014 [CITE NEEDED]).

The deformation half-width explains why the profiles constrain $b$ at all (Figure 14). With the derived recurrence intervals, $aT \approx 149$ km at Krafla and $\approx 16$ km at Asal, comparable to the extent of each network on either side of the rift. Each network therefore samples the transition from the near-rift transient to the far-field plateau $\pm V/2$, rather than the transient alone or the plateau alone, and through $T = b/V$ this transition fixes the opening.

The fits also expose what the model leaves out. At Krafla, the derived $T = 89$–$95$ yr is about a third of the $\sim 250$ yr between the 1724–1729 and 1975–1984 crises. At Asal, the fitted opening $b = 0.67$ m is about a third of the $\approx 1.9$ m measured across the rift after the 1978 event (Ruegg et al. 1979). The predicted early lengthening rate exceeds the trilateration rate by 27 per cent (Ruegg & Kasser 1987), and the profile retains an SW–NE asymmetry that a symmetric model cannot produce ($\chi^2/N \approx 21$). Each misfit points to an ingredient absent from the model: a single, infinitely thin dike in one elastic layer, symmetric about the axis, with no shallow magmatic source and strictly periodic recurrence. Smittarello et al. (2016) argue that magma supply rather than visco-elastic relaxation controls the post-diking velocity field at Asal, and our single-parameter description cannot separate the two without an independent constraint on the magmatic budget. The fitted $(a, b)$ are therefore effective values for a symmetric, periodic rift, and the viscosities derived from them inherit this simplification.

### 5.5 Irregular recurrence

The RSS assumes that intrusions recur at a fixed interval $T$, with a fixed opening $b$, always at the same place, and none of this holds in nature. Magmatic rifting is episodic; the $\sim 250$ yr gap between the Krafla crises is a single estimate from a sparse historical record, dike openings vary within and between sequences (Thordarson & Larsen 2007), and successive dikes need not open on the same plane. We

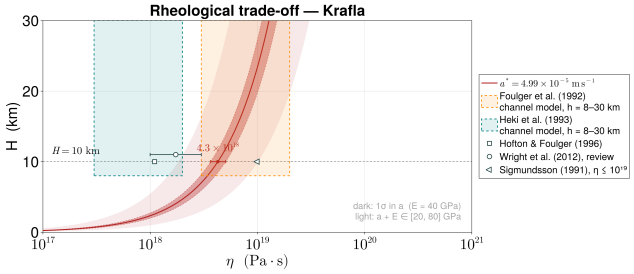



**Figure 12.** Rheological trade-off for Krafla, for the EV RSS best-fit $a = 4.99 \times 10^{-5}$ m s$^{-1}$ (Section 4.1). The panel shows the iso-$a$ contour in the $(H, \eta)$ plane at the best-fit value. Dark shading propagates the $1\sigma$ uncertainty on $a$ from the joint $\chi^2$ landscape at fixed $E = 40$ GPa; light shading additionally propagates $E$ over [20, 80] GPa (combined $1\sigma$ envelope). The $H = 10$ km dashed line marks the reference elastic thickness, and the filled circle with error bar the best fit at that thickness with its $1\sigma$ range at $E = 40$ GPa. Published estimates: dashed boxes, the channel-model viscosities of Foulger et al. (1992) and Heki et al. (1993) for an elastic layer of 8–30 km (viscosities of a thin channel, not of a half-space, so not directly comparable); open square, Hofton & Foulger (1996); open circle with error bar, Wright et al. (2012), drawn at $H = 11$ km for legibility; open triangle, the upper bound $\eta \leq 10^{19}$ Pa s of Sigmundsson (1991).

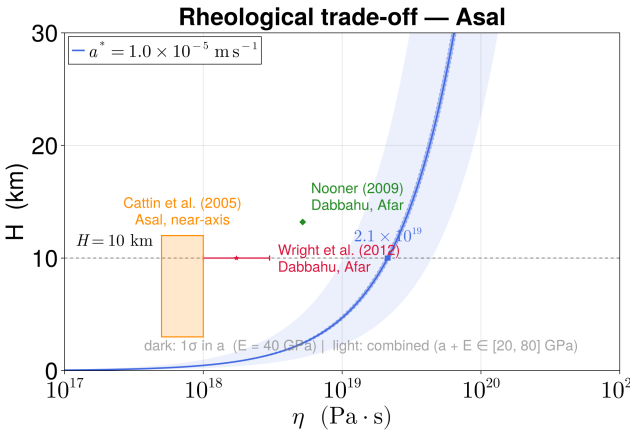



**Figure 13.** Rheological trade-off for Asal, for the EV RSS best-fit $a = 1.00 \times 10^{-5}$ m s$^{-1}$ (Section 4.2). The panel shows the iso-$a$ contour in the $(H, \eta)$ plane at the best-fit value. Dark shading propagates the $1\sigma$ uncertainty on $a$ from the joint $\chi^2$ landscape at fixed $E = 40$ GPa; light shading additionally propagates $E$ over [20, 80] GPa (combined $1\sigma$ envelope). The $H = 10$ km dashed line marks the reference elastic thickness; markers show published viscosity estimates: filled diamond, Nooner et al. 2009, and star with error bar, Wright et al. 2012, both for the Dabbahu segment of the same rift system; orange box, Cattin et al. 2005 — near-axis effective viscosity $\eta = 0.5$–$1 \times 10^{18}$ Pa s from a 2D thermo-mechanical FEM model of the first 6–8 yr post-rifting at Asal, with the seismogenic layer thickness $H = 3$–$12$ km from the same study. The Cattin et al. box lies well below the iso-$a$ curve, consistent with the local near-axis viscosity being lower than the laterally averaged bulk estimate from the 25-yr geodetic record.

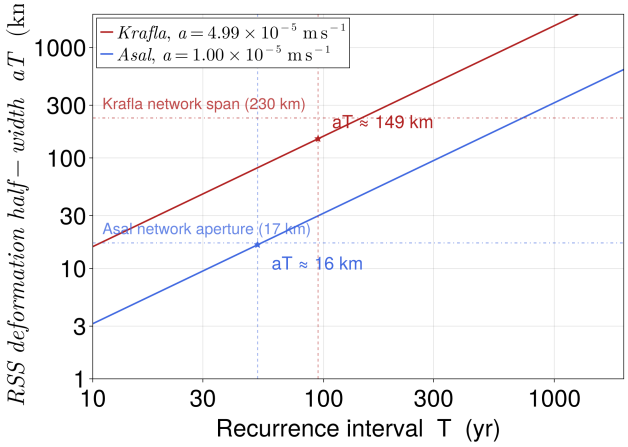



**Figure 14.** RSS deformation half-width $aT$ as a function of recurrence interval $T$ for Krafla (red, $a = 4.99 \times 10^{-5}$ m s$^{-1}$) and Asal (blue, $a = 1.00 \times 10^{-5}$ m s$^{-1}$), the EV RSS best fits. Vertical dashed lines mark the recurrence intervals derived from $T = b/V$ (95 yr and 52 yr); stars give the implied $aT$. Horizontal dash-dot lines mark the total span of each geodetic network used in this study: 230 km for Krafla (distance between outermost stations; Foulger et al. 1992) and 17 km for Asal (GPS profile extent; Vigny et al. 2007). The half-width $aT$ is comparable to the extent of each network on either side of the rift.

therefore tested how closely the RSS describes a rift whose timing, opening and dike position vary from cycle to cycle, keeping $T = b/V$ on average in every case. Each realisation sums the EV step velocities (15) of 300 past intrusions. We considered four cases (Figure 15): (a) openings $b_j$ drawn at random with $T$ fixed, a contrast in which the opening does not match the interval; (b) intervals $T_j$ drawn at random with each opening coupled to the preceding interval, $b_j = VT_j$; (c) the position $x_j$ of each intrusion drawn from a Normal distribution of standard deviation $\sigma_x = 0.1\,aT$, with periodic timing and opening; and (d) cases (b) and (c) combined. $T_j$ and $b_j$ follow Normal distributions with coefficient of variation $c_v = 0.30$, truncated at 5 per cent of the mean. Each case comprises 500 realisations at phase $\tau/T = 0.5$. A uniform distribution of $x_j$ with the same standard deviation gives bands within 10 per cent of the Normal case.

In all four cases the ensemble median follows the deterministic RSS to within 3 per cent of the far-field velocity $V/2$. For scatter in $b_j$ or $x_j$ alone, linearity fixes the ensemble mean: it equals the RSS for $b_j$, and the RSS convolved with the distribution of $x_j$ for $x_j$. Scatter in $T_j$ enters non-linearly through the ages of past intrusions, but the resulting bias is small at $c_v = 0.30$. The RSS therefore describes the typical velocity field of an irregular rift, and the bands measure how far a single realisation departs from it.

The three sources of scatter act at different distances from the axis, because each station integrates a different set of past intrusions. Near the axis, the contribution of an intrusion decays with its age $t$ as $1/t^2$ (Equation 15), so the most recent ones dominate; farther out, a station also feels every intrusion younger than about $|x|/a$. In units of $V/2$, the 10–90 per cent band of the contrast case (a) reaches 0.46 at $\xi = 0.3$ and still 0.20 at $\xi = 5$: when openings do not match the intervals, the plateau of a single realisation reflects the opening rate realised over the last few cycles. Coupling the openings to the intervals (b) removes most of that far-field spread (0.31 at $\xi = 0.3$, 0.06 at $\xi = 2$, 0.01 at $\xi = 5$), because the opening delivered over any time window then equals $V$ times that window. Scatter in dike position (c) acts only near the axis, where the band reaches 0.62 of $V/2$, and decays to 0.24 at $\xi = 0.3$ and below 0.01 at $\xi = 5$: far from the axis, a shift of the source by $\sigma_x$ barely changes the distance to the station. Combined, coupled timing and dike position (d) give a band of 0.6 at the axis that narrows to 0.18 at $\xi = 1$ and 0.02 at $\xi = 5$.

These results bear directly on the fits of Section 4. Both networks lie within one half-width of the axis, $|x| \lesssim 0.9\,aT$ at Krafla and $|x| \lesssim 0.6\,aT$ at Asal, where the bands of the coupled and dike-position cases are widest. The effect of dike position scales with $\sigma_x/(aT)$, so it matters more at Asal than at Krafla: a dike zone a few kilometres wide corresponds to $\sigma_x/(aT)$ of order 0.1 at Asal ($aT \approx 16$ km) but of order 0.01 at Krafla ($aT \approx 149$ km). The fitted $(a, b)$ therefore carry an uncertainty from cycle-to-cycle variability that the $\chi^2$ landscapes do not include. The bands of Figure 15 are computed at one phase and one level of scatter, so they indicate the size of this effect rather than quantify it for either site.

### 5.6 Perspectives

The RSS velocity depends on three combinations of the model parameters: the half-width $aT$, the far-field velocity $b/(2T) = V/2$, and the phase $\tau/T$ (Equation 23). Our fits derive the recurrence interval from the fitted opening and an independent rift opening rate, $T = b/V$ (Section 4). Records that span a larger fraction of a cycle, or snapshots repeated over decades, would sample the evolution of the profile with phase and so test this derived $T$ against geodesy. Propagating the stochastic spread of Section 5.5 into the inferred parameters, with ensembles conditioned on each site's phase and intrusion history, would complete the uncertainty budget.

Real rifts are segmented: dikes span finite along-strike lengths, and segments differ in recurrence history, plate thickness and heat flow. Because the only length scale of the model is the local product $aT$, each segment generates its own deformation halo, and to the extent that the segments respond linearly, the surface velocity is their superposition. The 1D model cannot capture along-strike dike propagation or the interaction between segments. Extending the RSS to the full 2D problem requires relaxing the plane-strain approximation while retaining the non-local Stokes coupling at the base of the plate.

## 6 CONCLUSIONS

Inter-rifting deformation at magmatic rifts can be described as the superposed relaxation of a long history of discrete dike intrusions in an elastic plate over a viscous half-space. In the limit of many intrusions, this superposition reaches a Recurrent Steady State with a closed form in the Digamma and log-Gamma functions. The RSS conserves the opening of each intrusion over one cycle, so far-field plate divergence follows from the model itself rather than from a locking

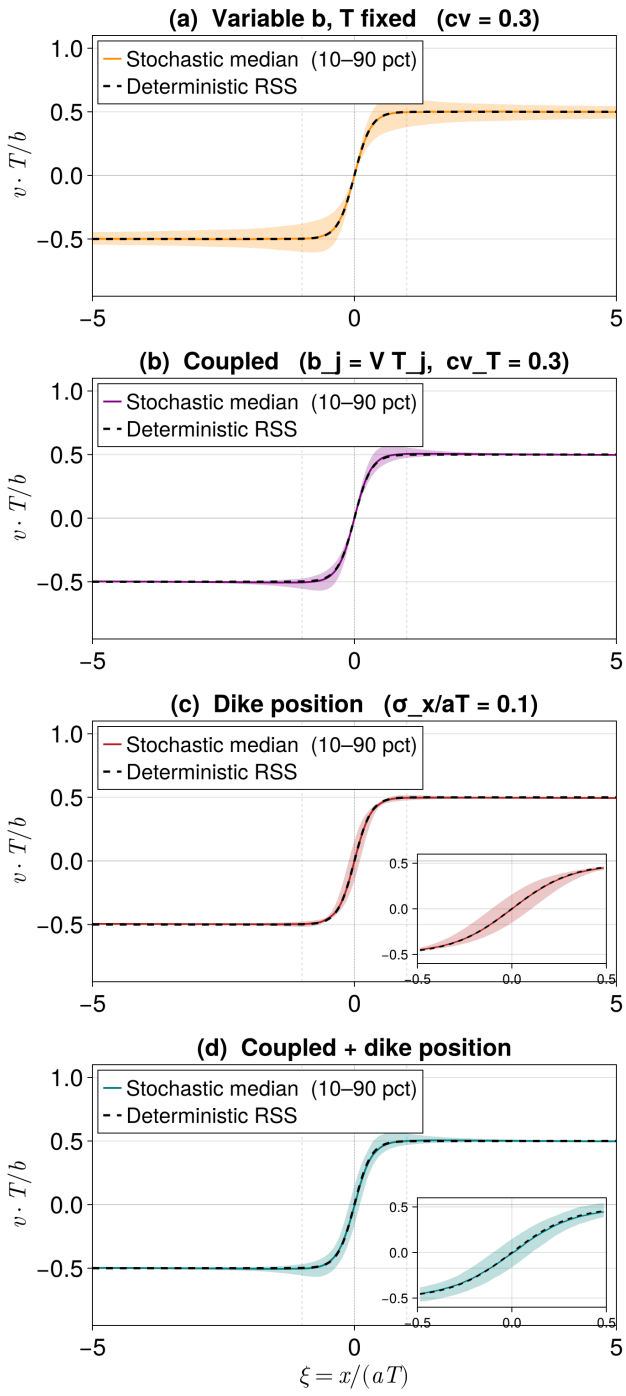



**Figure 15.** EV RSS velocity profile under stochastic recurrence ($a = T = b = 1$; phase $\tau/T = 0.5$; 300 past intrusions; 500 realisations; $T = b/V$ on average in every case). Velocity is normalised by $b/T$ and distance by $aT$. Shaded bands show the 10th–90th percentile range; coloured lines are ensemble medians; the black dashed line is the deterministic RSS (23); vertical dashed lines mark $\xi = \pm 1$. (a) Contrast: openings $b_j$ random ($c_v = 0.30$) with $T$ fixed. (b) Intervals $T_j$ random ($c_v = 0.30$) with each opening coupled to the preceding interval, $b_j = VT_j$. (c) Dike position $x_j$ random, Normal with $\sigma_x = 0.1\,aT$, periodic timing and opening. (d) Cases (b) and (c) combined. Insets in (c) and (d) enlarge $|\xi| \le 0.5$. In all four cases the ensemble median stays within 3 per cent of $V/2$ of the deterministic RSS.

depth, a prescribed domain width or a boundary condition, and its only length scale is the deformation half-width $aT$. A steady velocity far from the rift is therefore the memory of intrusions that occurred many cycles earlier, not evidence of continuous creep at depth. This memory also sets the validity of the RSS: a station at distance $|x|$ is described by it only if the rift has hosted more than about $13\,|x|/(aT)$ intrusions.

The coupling of the plate to the half-space through the Stokes kernel turns post-diking relaxation into a fractional diffusion governed by the half-Laplacian. Deformation then spreads from the rift at the constant speed $a$ with a $1/x$ tail, instead of the decelerating front and Gaussian tail of Fickian diffusion. The Fickian model of Foulger et al. (1992) is the same relaxation evaluated at a single diffusion time, whereas the EV solution averages Fickian profiles over a heavy-tailed distribution of diffusion times. At Krafla, this difference is visible in the data: under the same conditions, the EV RSS fits the two post-rifting profiles with $\chi^2_r = 1.61$, against 4.90 for the diffusion model.

A Maxwell half-space changes the response only within the Maxwell length $\ell_M$ of the axis and during the first few relaxation times $\tau_R$. With the plate and the half-space sharing one elastic modulus, $\ell_M = H/(1-\nu) \approx 1.3\,H$, so a geodetic record can distinguish the two rheologies only if it starts early in units of $\tau_R$ and has stations close to the axis. The Krafla record, which starts $3.4\,\tau_R$ after the onset, favours the viscous half-space only modestly ($\Delta$AIC $= +7.1$); the Asal record, which starts at $0.2\,\tau_R$, favours it decisively ($\Delta$AIC $= +70.3$). The choice of rheology also changes the inferred viscosity, by a factor of up to seven at Asal.

Tying the recurrence interval to an independent rift opening rate, $T = b/V$, leaves two free parameters, $a$ and $b$ (plus a reference-frame offset at Krafla), which the Krafla and Asal records both constrain. For $H = 10$ km and $E = 40$ GPa, the EV RSS implies $\eta \approx 4.3 \times 10^{18}$ Pa s at Krafla and $\eta \approx 2.1 \times 10^{19}$ Pa s at Asal, with an error budget dominated by the elastic modulus rather than by the fit. The remaining misfits, a recurrence interval shorter than the Krafla eruptive history and an opening and a profile asymmetry at Asal that the model does not reproduce, point to the simplifications of the model: a single, infinitely thin dike in one elastic layer, symmetric about the axis, with no shallow magmatic source and strictly periodic recurrence.

Irregular recurrence spreads individual velocity profiles around the RSS with little bias: with 30 per cent scatter in recurrence interval and opening, and dike positions scattered over $0.1\,aT$, the ensemble median stays within 3 per cent of the deterministic RSS. The spread is largest within about one half-width of the axis, where the velocity reflects the last few intrusions. Scatter in dike position acts only within a few $\sigma_x$ of the axis, and the far-field spread nearly vanishes when each opening scales with the preceding interval, as $T = b/V$ requires. Extending the framework to segmented rifts and to the full along-strike 2D geometry, while retaining the non-local coupling of the plate to the half-space, is the next step toward a system-scale model of strain accumulation at magmatic plate boundaries.

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## Data Availability

The geodetic data used in this study are from published sources cited in the text. The analytical model code and inversion scripts are available at [PLACEHOLDER — repository DOI or URL upon acceptance].

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## APPENDIX A: FOURIER TRANSFORM OF THE STEP INITIAL CONDITION

The Fourier transform of the initial condition (11) is $\hat{u}(k,0) = b\int_{-\infty}^{0} e^{-ikx}\,dx$, which for real $k$ is treated in the distributional sense by adding a convergence regulator $\varepsilon > 0$:

$$\int_{-\infty}^{0} e^{-ikx}\,dx = \lim_{\varepsilon\to 0}\int_{-\infty}^{0} e^{(-ik+\varepsilon)x}\,dx = \lim_{\varepsilon\to 0}\frac{1}{\varepsilon - ik}.$$

By the Sokhotski–Plemelj theorem (Ablowitz & Fokas 2003),

$$\lim_{\varepsilon\to 0^+}\frac{1}{x \pm i\varepsilon} = \mathcal{P}\left(\frac{1}{x}\right) \mp i\pi\delta(x), \tag{A.1}$$

where $\mathcal{P}$ denotes the Cauchy principal value. Multiplying numerator and denominator by $i$ recasts the regulator as $+i\varepsilon$,

$$\frac{1}{\varepsilon - ik} = \frac{i}{i\varepsilon + k} = \frac{i}{k + i\varepsilon},$$

so that (A.1) applies directly with $x = k$ and the upper sign:

$$\begin{aligned}\lim_{\varepsilon\to 0^+}\frac{1}{\varepsilon - ik} &= \lim_{\varepsilon\to 0^+}\frac{i}{k + i\varepsilon}\\ &= i\left[\mathcal{P}\left(\frac{1}{k}\right) - i\pi\delta(k)\right]\\ &= \pi\delta(k) + \frac{i}{k},\end{aligned} \tag{A.2}$$

where the principal-value symbol on $1/k$ is dropped, as is standard, since it is understood implicitly whenever $1/k$ appears next to a $\delta(k)$ term. This gives (12).

The inverse transform of the evolved spectrum (13) splits into two contributions, $u(x,t) = I_1 + I_2$. The delta-function term gives a constant background $I_1 = b/2$; for $I_2$, the parity of $\sin(kx)/k$ yields

$$I_2 = -\frac{b}{\pi}\int_0^{\infty}\frac{\sin(kx)}{k}\,e^{-akt}\,dk = -\frac{b}{\pi}\arctan\!\left(\frac{x}{at}\right), \tag{A.3}$$

a standard Dirichlet-type integral. With the complementary arctangent identity $\arctan(z) + \arctan(1/z) = \pi/2$, the displacement counted from the instant after the intrusion is the step solution (14).

## APPENDIX B: DIGAMMA REPRESENTATION OF THE RSS VELOCITY

Each term of the RSS series (22) splits into partial fractions,

$$\frac{x}{x^2 + a^2(\tau + jT)^2} = \frac{1}{2i}\left[\frac{1}{a(\tau + jT) - ix} - \frac{1}{a(\tau + jT) + ix}\right].$$

Factoring $aT$ from each denominator and substituting into (22), the sum takes the form of a difference of two Digamma series. With the representation $\psi(z) = -\gamma + \sum_{j=0}^{\infty}[1/(j+1) - 1/(j+z)]$ and $z = \tau/T \pm ix/(aT)$, the Euler–Mascheroni constant $\gamma$ and the $1/(j+1)$ terms cancel when the two conjugate series are differenced. With $Z = \tau/T + ix/(aT)$ and the identity $\psi(Z) - \psi(\bar{Z}) = 2i\,\mathrm{Im}\{\psi(Z)\}$, the series reduces to (23).

## APPENDIX C: CORRESPONDENCE PRINCIPLE FOR THE MAXWELL HALF-SPACE

For deviatoric stress $\tau_{ij}$ and strain $\varepsilon$, the 1D constitutive relation of a Maxwell body is

$$\dot{\varepsilon} = \frac{\dot{\tau}}{2\mu_m} + \frac{\tau}{2\eta}. \tag{C.1}$$

Applying the Laplace transform $\tilde{f}(s) = \mathcal{L}\{f(t)\}$ to (C.1) with zero initial conditions gives

$$s\tilde{\varepsilon} = \tilde{\tau}\left(\frac{s}{2\mu_m} + \frac{1}{2\eta}\right). \tag{C.2}$$

Solving for stress in terms of the strain rate $s\tilde{\varepsilon}$ gives $\tilde{\tau} = 2\tilde{\eta}(s)\,(s\tilde{\varepsilon})$, with the effective viscosity $\tilde{\eta}(s)$ of (26). Laplace-transforming the Stokes kernel (3) and replacing $\eta$ by $\tilde{\eta}(s)$ gives

$$\tilde{\sigma}_{xz}^{\mathrm{BOT}}(x,s) = \int_{-\infty}^{+\infty}\frac{2\tilde{\eta}(s)}{\pi(x-x')^2}\left[s\tilde{u}(x',s)\right]dx',$$

or, with (26),

$$(1 + \tau_R s)\,\tilde{\sigma}_{xz}^{\mathrm{BOT}}(x,s) = \int_{-\infty}^{+\infty}\frac{2\eta}{\pi(x-x')^2}\left[s\tilde{u}(x',s)\right]dx'. \tag{C.3}$$

Since multiplication by $s$ in the Laplace domain corresponds to $\partial/\partial t$, (C.3) inverts to the visco-elastic Stokes kernel (27).

## APPENDIX D: NUMERICAL EVALUATION OF THE EVE INTEGRALS

The EVE step integrals (33) and (34) are evaluated with adaptive Gauss–Kronrod quadrature (Laurie 1997), with

integration breakpoints at the successive zeros of $\sin(kx)$, $k = n\pi/x$ ($n = 1, 2, \ldots$), which partition the oscillatory integrand into well-behaved sub-intervals. The upper truncation wavenumber is set adaptively so that the exponential factor is negligible at the cutoff.

The EVE RSS integrals (40) and (41) contain factors that tend to finite, non-zero limits as $k \to \infty$, so their integrands decay slowly. For the fits of Section 4, the large-$k$ limit of each integrand is subtracted and integrated in closed form with the Dirichlet integral, which also gives the axis jump (43). The remainder decays as $1/k^2$ on the scale $k \sim 1/(a\tau_R)$ and is integrated with the same segmented quadrature up to the fixed cutoff $10^4/(a\tau_R)$.